

Archived local-eligibility parameter landscapes

Full forward sweeps of all 12 learnable parameters in the frozen pre-dPSC July 8/9 implementation

For the eight PSTH-loss parameters, 100/103 qualified interior points were stable green and 3 were stable red; 220 additional points were inconclusive. For the four CV-loss refractory parameters, 31/31 qualified points were stable green and 0 were stable red; 110 were inconclusive and 0 had an undefined CV stencil.

This is an exploratory directional-consistency check between the archived eligibility-based update signal and a coarse sampled hard-spike loss curve. It is not proof of an exact derivative or parameter identifiability. In particular, the archived refractory rule multiplies the CV residual by a refractory accumulator but does not explicitly differentiate CV through interspike intervals and spike times. Every point is a full 2.9801 s simulation; no Adam update is applied.

Snapshot	Saved run	Cells	Seeds	Sweep rows	Forward cell-runs
pre-dPSC local snapshot (Jul 9)	100_epoch_post_strf_fix_4.mat	1, 2, 7	123, 456, 789	159	1431

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Methods and interpretation

One parameter at a time

For every parameter, a broad grid spanning its legal or practically relevant range was evaluated. The other 11 parameters were held at each cell's selected final tracker value from 100_epoch_post_strf_fix_4.mat.

Fixed cell-specific reference solutions

Cells 1, 2, and 7 use zero-based saved-output batches 6, 3, and 3, respectively. Those batches were selected in-sample from the old run by minimum saved PSTH SSE. The final tracker slice is used because it is the parameter state that generated the saved raster; the post-final-Adam continuation checkpoint is intentionally not used.

Common random numbers

Within a stochastic seed, all parameter values receive identical underlying uniforms for onset spikes, offset spikes, spontaneous spikes, and the probabilistic output-spike gate. This makes neighboring loss differences much less noisy than independent reruns. Results are averaged over seeds 123, 456, 789.

Raw color and qualified criterion

Every retained plotted point has the requested raw face color: green when the mean archived eligibility-based update signal and the adjacent-point sampled slope have the same nonzero sign, red otherwise. The stricter headline excludes endpoints, zero or turning secants, undefined CV stencils, and any point without unanimous seed-level classification. A qualified point is stable green only when every seed has the same nonzero update-signal sign and the same nonzero slope sign and those signs match; stable red requires unanimous but opposite signs. This is descriptive unanimity with 3 seeds, not a confidence interval.

Two historical objectives

STRF gain, STRF alpha, output adaptation, and the five conductances use summed PSTH SSE. The four refractory parameters use the CV squared-error objective because that is how the archived Loss_handler routes their update signals.

Plot legend: green/red face = the raw sign comparison requested for every point; gray open ring = inconclusive for the qualified analysis; gray X = an undefined CV stencil; purple dashed line = the parameter value that generated the selected saved raster. The lower panels normalize both signals by the largest absolute value on that cell/parameter curve, so they compare sign and shape rather than units.

Final tracker values held fixed during each sweep

Parameter	Cell 1 (batch 6)	Cell 2 (batch 3)	Cell 7 (batch 3)
STRF gain	0.015169	0.014654	0.0069216
STRF alpha (ms)	25.938	29.201	18.432
Output adaptation	0.0052543	0.0037856	0.0043002
on -> ron gSYN	0.016436	0.0049576	0.011607
off -> ron gSYN	0.001	0.0013098	0.0034953
on -> sonoff gSYN	0.005271	0.0010005	0.02881
off -> sonoff gSYN	0.001	0.001	0.002133
sonoff -> ron gSYN	0.057351	0.030888	0.035884
Absolute refractory period (ms)	0	3.6539	0
Relative refractory a	8.796	3.245	8.1406
Relative refractory b	0	8.555	1.1437
Relative refractory c	1	1	1

The raw sweep includes each saved-output parameter value itself. A baseline point is omitted from the slope/plot analysis when it is exactly duplicated or lies within 2% of the typical intentional grid spacing from a grid point; every such run remains in raw_results.csv.

Qualified directional-consistency summary

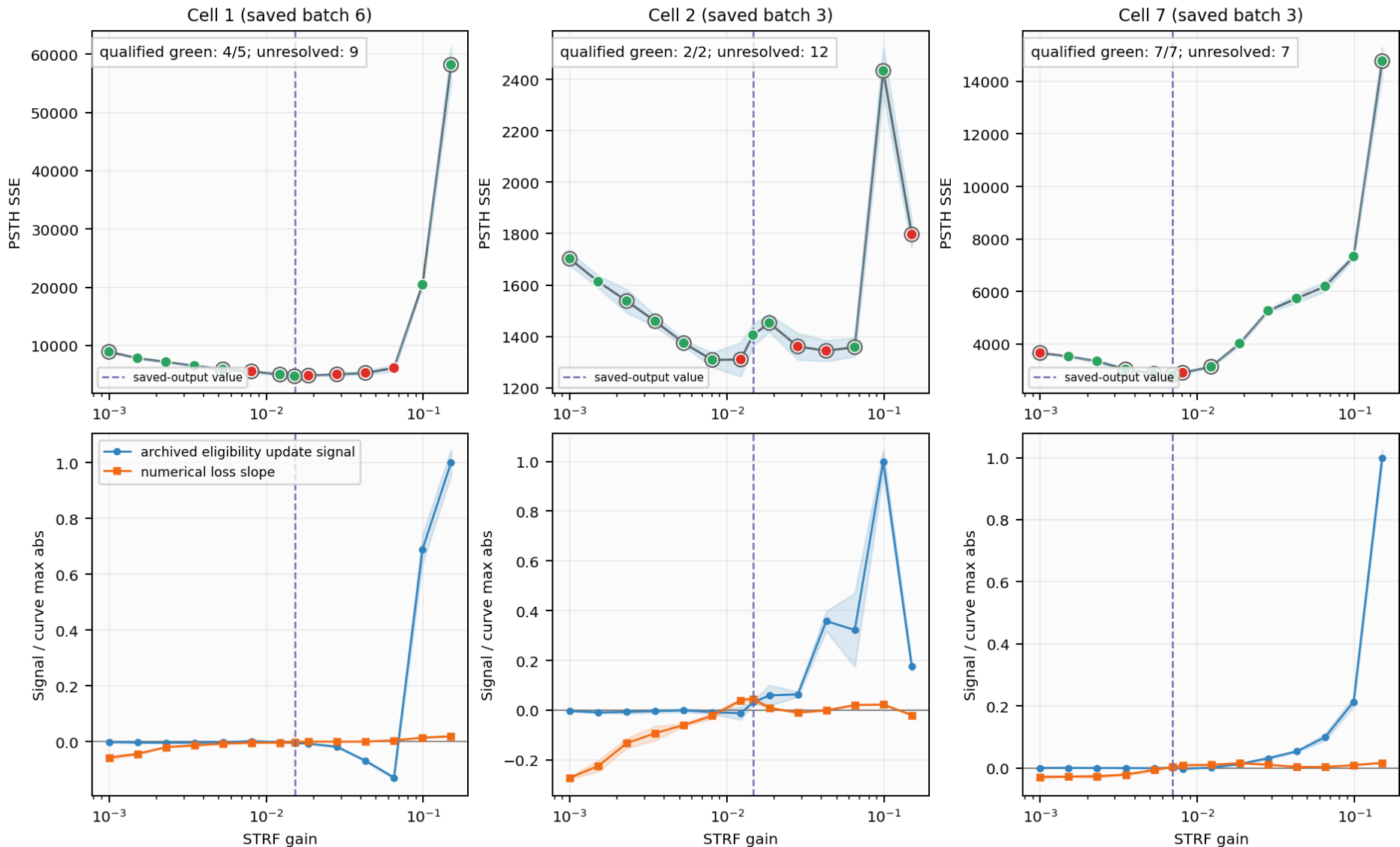
Parameter	Objective	Stable G / R	Qualified %	Inconcl.	Invalid CV	Raw G / all	Cell 1	Cell 2	Cell 7
STRF gain	PSTH SSE	13 / 1	92.9%	28	0	31 / 42	80.0%	100.0%	100.0%
STRF alpha (ms)	PSTH SSE	8 / 2	80.0%	31	0	17 / 41	66.7%	100.0%	100.0%
Output adaptation	PSTH SSE	22 / 0	100.0%	14	0	33 / 36	100.0%	100.0%	100.0%
on -> ron gSYN	PSTH SSE	30 / 0	100.0%	12	0	42 / 42	100.0%	100.0%	100.0%
off -> ron gSYN	PSTH SSE	27 / 0	100.0%	13	0	36 / 40	100.0%	100.0%	100.0%
on -> sonoff gSYN	PSTH SSE	0 / 0	N/A	40	0	18 / 40	N/A	N/A	N/A
off -> sonoff gSYN	PSTH SSE	0 / 0	N/A	40	0	23 / 40	N/A	N/A	N/A
sonoff -> ron gSYN	PSTH SSE	0 / 0	N/A	42	0	19 / 42	N/A	N/A	N/A
Absolute refractory period (ms)	CV squared error	22 / 0	100.0%	12	0	33 / 34	100.0%	100.0%	100.0%
Relative refractory a	CV squared error	2 / 0	100.0%	37	0	10 / 39	N/A	100.0%	N/A
Relative refractory b	CV squared error	0 / 0	N/A	35	0	7 / 35	N/A	N/A	N/A
Relative refractory c	CV squared error	7 / 0	100.0%	26	0	16 / 33	100.0%	100.0%	100.0%

Qualified percentages use only stable-green plus stable-red points. Inconclusive and invalid-CV points are shown separately and excluded from the denominator. Raw G/all preserves the user's requested binary face-color comparison over every plotted point.

STRF gain

Objective: PSTH SSE. Qualified stable green/red: 13/1; qualified green fraction 92.9%. Inconclusive: 28; invalid CV: 0; raw green/all: 31/42.

STRF gain: archived local eligibility versus sampled loss slope



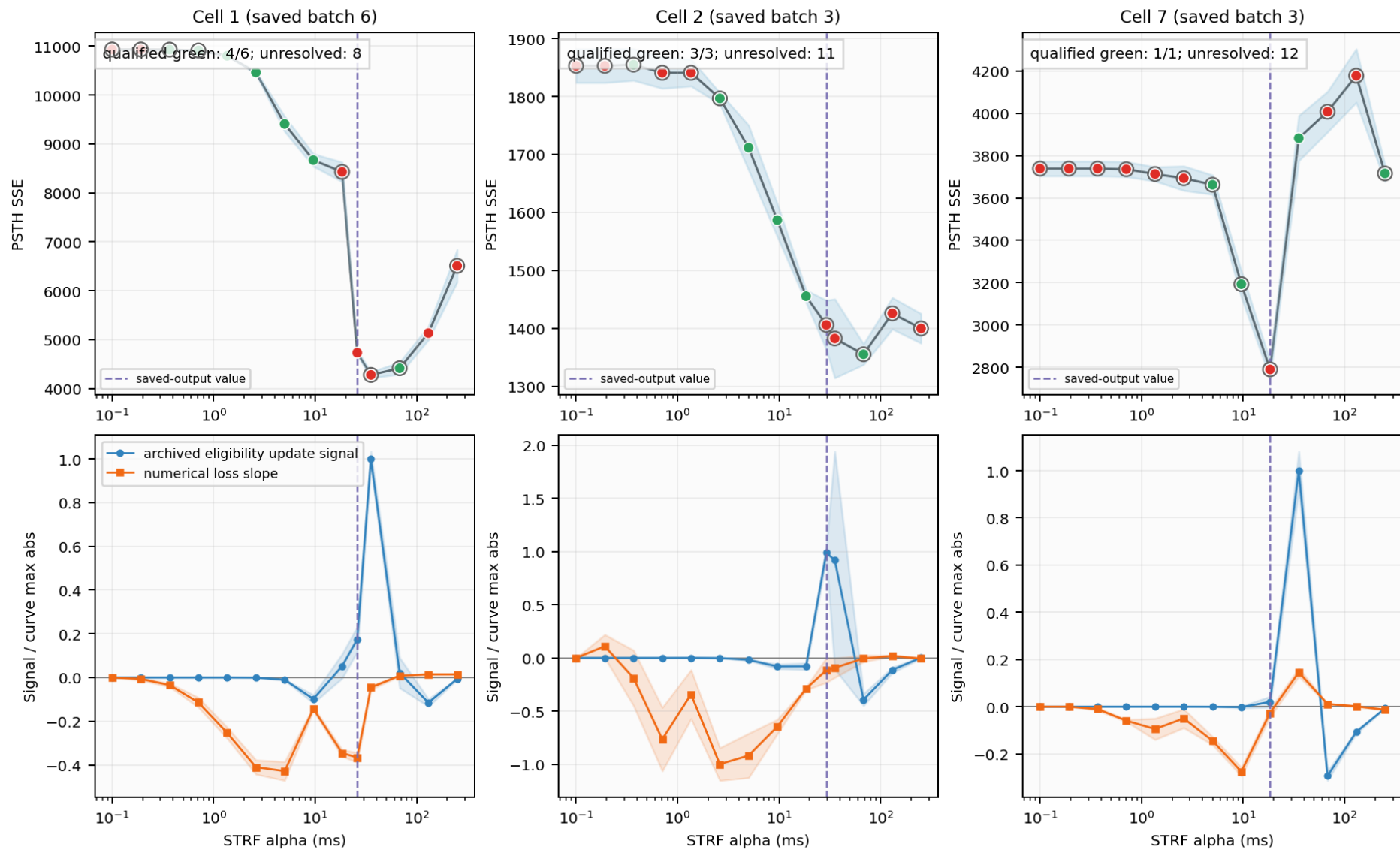
STRF gain: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.001	8912	-3.931e+04	-2.1136e+06	12.34	GREEN	INCONCLUSIVE	100.0%
1	0.0015182	7816.7	-1.0067e+05	-1.6010e+06	17.59	GREEN	STABLE GREEN	100.0%
1	0.0023051	7169.3	-1.2413e+05	-7.1409e+05	23.25	GREEN	STABLE GREEN	100.0%
1	0.0034996	6513.3	-1.0877e+05	-4.6464e+05	29.14	GREEN	STABLE GREEN	100.0%
1	0.0053133	5903.3	-6.425e+04	-2.5181e+05	34.69	GREEN	INCONCLUSIVE	100.0%
1	0.0080669	5563.3	8.63e+04	-1.2545e+05	39.94	RED	INCONCLUSIVE	100.0%
1	0.012247	5026.3	-5.429e+04	-1.1255e+05	43.41	GREEN	INCONCLUSIVE	100.0%
1	0.015169	4730	-1.5127e+05	-4.011e+04	44.66	GREEN	INCONCLUSIVE	100.0%
1	0.018595	4839	-2.4830e+05	2.924e+04	45.22	RED	INCONCLUSIVE	100.0%
1	0.028231	5051	-6.5807e+05	2.002e+04	46.04	RED	INCONCLUSIVE	100.0%
1	0.042862	5300	-2.5242e+06	2.534e+04	46.01	RED	INCONCLUSIVE	100.0%
1	0.065074	6143.3	-4.7662e+06	1.9110e+05	45.09	RED	STABLE RED	100.0%
1	0.098798	20429	2.5571e+07	5.4821e+05	47.87	GREEN	STABLE GREEN	100.0%
1	0.15	58185	3.7075e+07	7.3739e+05	41.37	GREEN	INCONCLUSIVE	100.0%
2	0.001	1702.3	-2087	-1.7173e+05	6.644	GREEN	INCONCLUSIVE	100.0%
2	0.0015182	1613.3	-5818	-1.4156e+05	7.908	GREEN	STABLE GREEN	100.0%
2	0.0023051	1538	-4328	-8.387e+04	9.519	GREEN	INCONCLUSIVE	100.0%
2	0.0034996	1459.3	-2241	-5.831e+04	11.46	GREEN	INCONCLUSIVE	100.0%
2	0.0053133	1374.3	-230.6	-3.768e+04	13.42	GREEN	INCONCLUSIVE	100.0%
2	0.0080669	1309	-4890	-1.415e+04	15.26	GREEN	INCONCLUSIVE	100.0%
2	0.012247	1310.7	-7724	2.52e+04	16.79	RED	INCONCLUSIVE	100.0%
2	0.014654	1405.7	1.979e+04	2.904e+04	17.61	GREEN	STABLE GREEN	100.0%
2	0.018595	1452.7	3.742e+04	5713	18.15	GREEN	INCONCLUSIVE	100.0%
2	0.028231	1361.3	4.025e+04	-6176	18.78	RED	INCONCLUSIVE	100.0%
2	0.042862	1344.3	2.2620e+05	-444.3	19.22	RED	INCONCLUSIVE	100.0%
2	0.065074	1358.7	2.0351e+05	1.304e+04	19.56	GREEN	INCONCLUSIVE	100.0%
2	0.098798	2433	6.3143e+05	1.427e+04	16.73	GREEN	INCONCLUSIVE	100.0%
2	0.15	1797	1.1144e+05	-1.242e+04	9.552	RED	INCONCLUSIVE	100.0%
7	0.001	3656.3	624.1	-2.6050e+05	3.412	RED	INCONCLUSIVE	100.0%
7	0.0015182	3521.3	-454.8	-2.4874e+05	4.608	GREEN	STABLE GREEN	100.0%
7	0.0023051	3339.7	-1768	-2.4181e+05	6.611	GREEN	STABLE GREEN	100.0%
7	0.0034996	3031	-3690	-1.9140e+05	9.295	GREEN	INCONCLUSIVE	100.0%
7	0.0053133	2868.3	-7831	-6.072e+04	12.81	GREEN	INCONCLUSIVE	100.0%
7	0.0069216	2812	915.4	3.167e+04	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.0080669	2902.7	-2.371e+04	7.389e+04	16.44	RED	INCONCLUSIVE	100.0%
7	0.012247	3131	5658	8.83e+04	19.92	GREEN	INCONCLUSIVE	100.0%
7	0.018595	4016	1.1047e+05	1.3549e+05	23.06	GREEN	STABLE GREEN	100.0%
7	0.028231	5264	2.7731e+05	9.083e+04	26.62	GREEN	STABLE GREEN	100.0%
7	0.042862	5733.7	4.7289e+05	2.758e+04	28.82	GREEN	STABLE GREEN	100.0%
7	0.065074	6194	8.8164e+05	2.589e+04	31.59	GREEN	STABLE GREEN	100.0%
7	0.098798	7331.3	1.8958e+06	7.805e+04	34.5	GREEN	STABLE GREEN	100.0%
7	0.15	14773	8.8912e+06	1.4535e+05	34.54	GREEN	INCONCLUSIVE	100.0%

STRF alpha (ms)

Objective: PSTH SSE. Qualified stable green/red: 8/2; qualified green fraction 80.0%. Inconclusive: 31; invalid CV: 0; raw green/all: 17/41.

STRF alpha (ms): archived local eligibility versus sampled loss slope



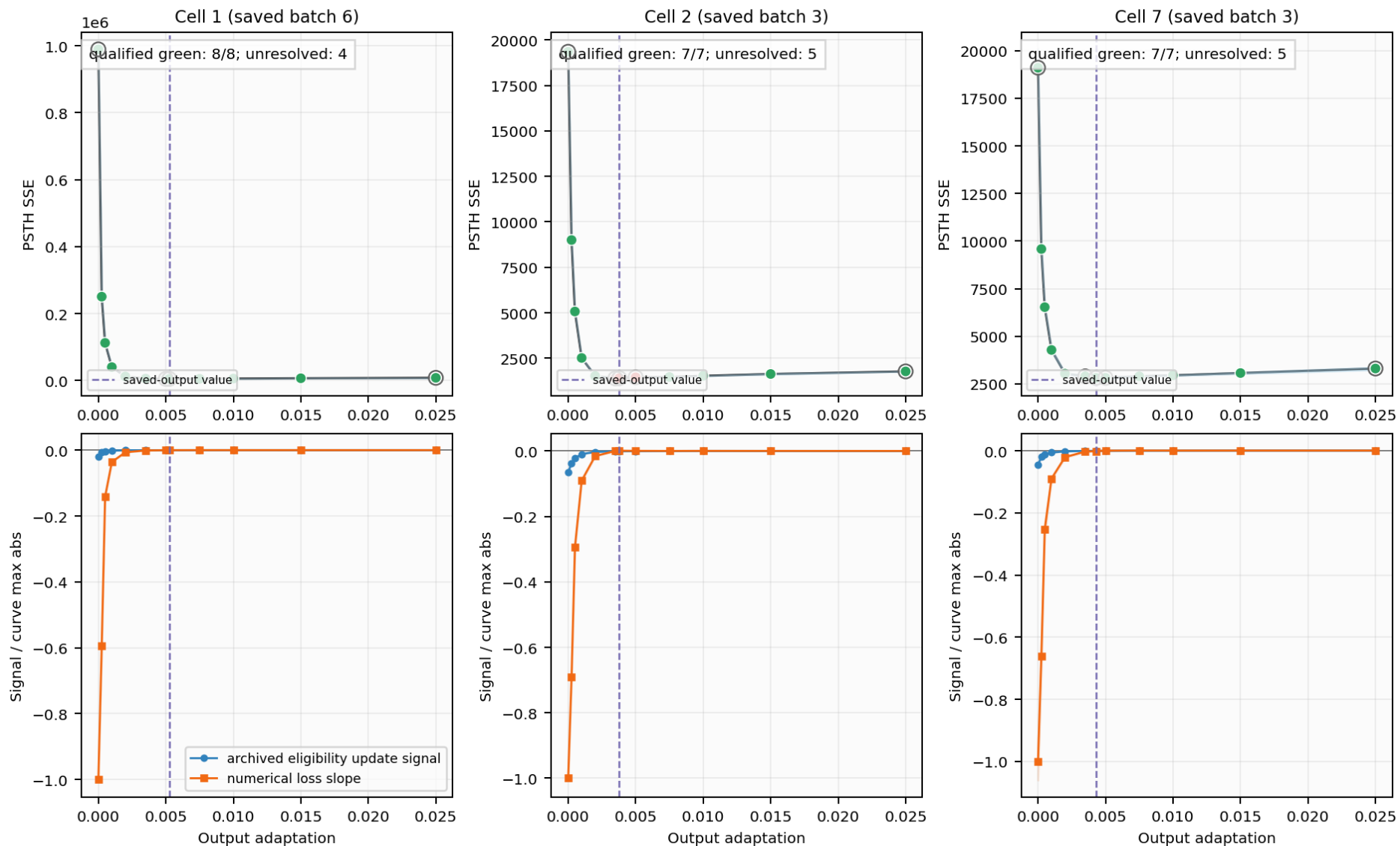
STRF alpha (ms): numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.1	10931	0	0	3.255	RED	INCONCLUSIVE	100.0%
1	0.19194	10931	0	-5.176	3.255	RED	INCONCLUSIVE	100.0%
1	0.3684	10928	-1.4686e-12	-27.8	3.255	GREEN	INCONCLUSIVE	100.0%
1	0.70711	10910	-2.5492e-05	-91.72	3.333	GREEN	INCONCLUSIVE	100.0%
1	1.3572	10801	-0.003468	-203.6	3.59	GREEN	STABLE GREEN	100.0%
1	2.605	10461	-0.4123	-330.1	4.731	GREEN	STABLE GREEN	100.0%
1	5	9406	-8.205	-344.7	8.691	GREEN	STABLE GREEN	100.0%
1	9.5969	8667.3	-78.91	-115	15.96	GREEN	STABLE GREEN	100.0%
1	18.42	8426	42.09	-278.1	28.38	RED	INCONCLUSIVE	100.0%
1	25.938	4730	140.4	-294.8	44.66	RED	STABLE RED	100.0%
1	35.355	4275.7	805.2	-36.46	45.52	RED	INCONCLUSIVE	100.0%
1	67.86	4412	17.28	6.703	44.35	GREEN	INCONCLUSIVE	100.0%
1	130.25	5130.7	-92.62	11.51	44.22	RED	STABLE RED	100.0%
1	250	6507	-5.102	11.49	36.84	RED	INCONCLUSIVE	100.0%
2	0.1	1853	0	0	4.273	RED	INCONCLUSIVE	100.0%
2	0.19194	1853	0	3.882	4.273	RED	INCONCLUSIVE	100.0%
2	0.3684	1855	-7.9588e-14	-6.707	4.273	GREEN	INCONCLUSIVE	100.0%
2	0.70711	1841	1.2643e-04	-27.18	4.362	RED	INCONCLUSIVE	100.0%
2	1.3572	1841	0.02044	-12.17	4.496	RED	INCONCLUSIVE	100.0%
2	2.605	1796.7	-0.0412	-35.52	4.933	GREEN	INCONCLUSIVE	100.0%
2	5	1711.7	-0.6384	-32.65	5.839	GREEN	STABLE GREEN	100.0%
2	9.5969	1586.7	-2.862	-22.98	7.953	GREEN	STABLE GREEN	100.0%
2	18.42	1455.3	-2.814	-10.26	12.05	GREEN	STABLE GREEN	100.0%
2	29.201	1405.7	35.14	-4.088	17.61	RED	INCONCLUSIVE	100.0%
2	35.355	1382.3	32.61	-3.322	18.16	RED	INCONCLUSIVE	100.0%
2	67.86	1355	-13.96	-0.1649	18.41	GREEN	INCONCLUSIVE	100.0%
2	130.25	1425.7	-4.019	0.6703	17.05	RED	INCONCLUSIVE	100.0%
2	250	1399.7	0.1033	-0.2171	12.35	RED	INCONCLUSIVE	100.0%
7	0.1	3738.7	0	0	2.796	RED	INCONCLUSIVE	100.0%
7	0.19194	3738.7	0	1.5158e-12	2.796	RED	INCONCLUSIVE	100.0%
7	0.3684	3738.7	0	-3.034	2.796	RED	INCONCLUSIVE	100.0%
7	0.70711	3735.7	2.3928e-08	-17.77	2.808	RED	INCONCLUSIVE	100.0%
7	1.3572	3713	3.2289e-04	-28.41	2.897	RED	INCONCLUSIVE	100.0%
7	2.605	3693	0.01047	-14.92	3.098	RED	INCONCLUSIVE	100.0%
7	5	3662.3	-0.06294	-43.42	3.658	GREEN	INCONCLUSIVE	100.0%
7	9.5969	3192.7	-0.5506	-82.83	6.823	GREEN	INCONCLUSIVE	100.0%
7	18.42	2789.3	6.158	-7.94	15.03	RED	INCONCLUSIVE	100.0%
7	35.355	3882.7	300.5	43.76	31.77	GREEN	STABLE GREEN	100.0%
7	67.86	4007.7	-88.23	3.465	32.83	RED	INCONCLUSIVE	100.0%
7	130.25	4178.3	-32.17	0.475	28.58	RED	INCONCLUSIVE	100.0%
7	250	3715.7	-2.089	-3.864	16.7	GREEN	INCONCLUSIVE	100.0%

Output adaptation

Objective: PSTH SSE. Qualified stable green/red: 22/0; qualified green fraction 100.0%. Inconclusive: 14; invalid CV: 0; raw green/all: 33/36.

Output adaptation: archived local eligibility versus sampled loss slope



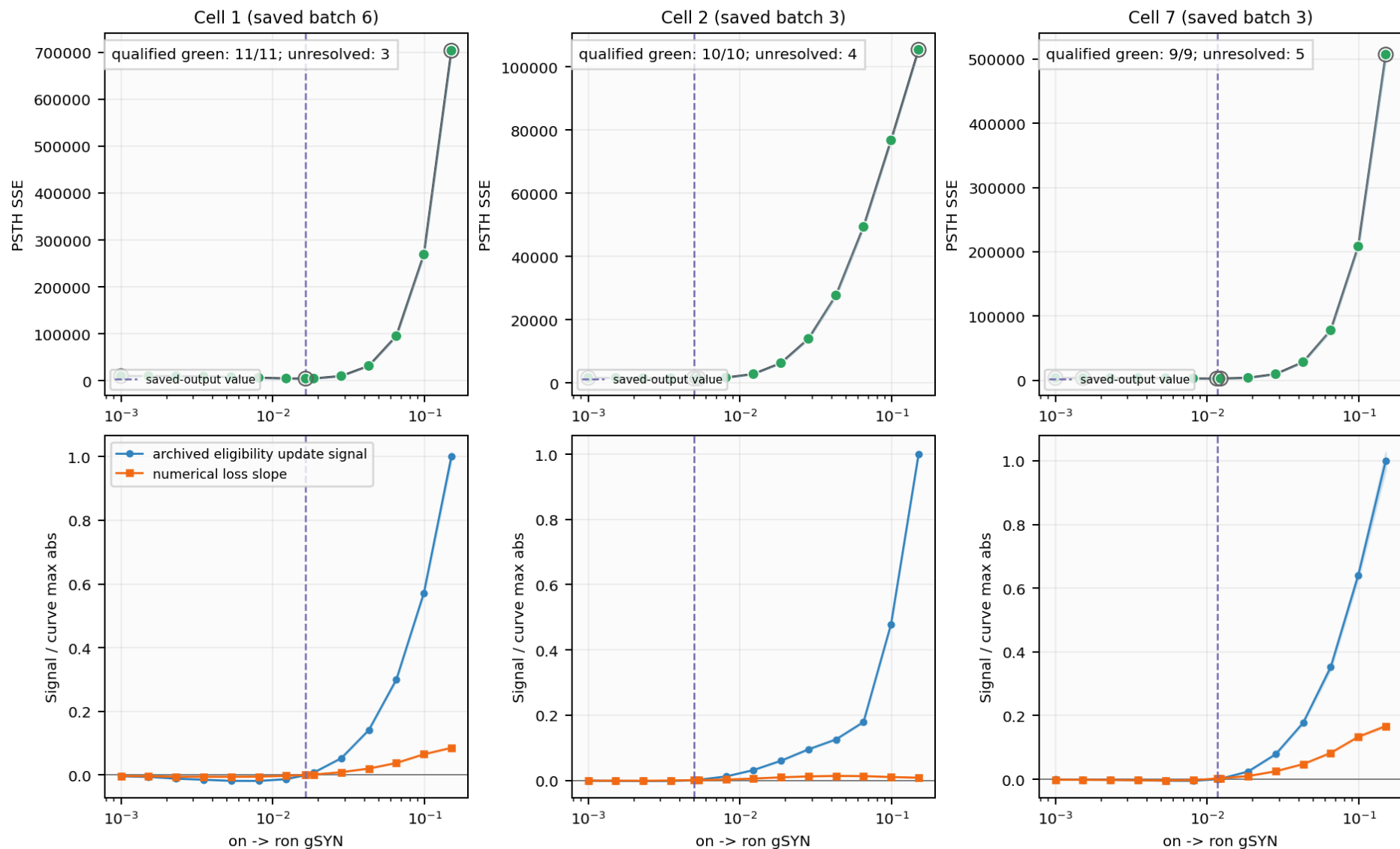
Output adaptation: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0	9.89081e+05	-5.6461e+07	-2.9545e+09	598.1	GREEN	INCONCLUSIVE	100.0%
1	2.50000e-04	2.50467e+05	-1.8020e+07	-1.7535e+09	318	GREEN	STABLE GREEN	100.0%
1	5.00000e-04	1.12333e+05	-9.0613e+06	-4.1656e+08	224.7	GREEN	STABLE GREEN	100.0%
1	0.001	40034	-3.6057e+06	-1.0574e+08	147.2	GREEN	STABLE GREEN	100.0%
1	0.002	12023	-1.0909e+06	-1.8524e+07	91.69	GREEN	STABLE GREEN	100.0%
1	0.0035	5581.7	-2.7058e+05	-2.4148e+06	60.79	GREEN	STABLE GREEN	100.0%
1	0.005	4778.7	-3.445e+04	-2.4124e+05	46.45	GREEN	INCONCLUSIVE	100.0%
1	0.0052543	4730	-3448	-1.5749e+05	44.66	GREEN	INCONCLUSIVE	100.0%
1	0.0075	5048.7	1.0301e+05	2.0605e+05	33.9	GREEN	STABLE GREEN	100.0%
1	0.01	5742.3	1.3225e+05	2.5007e+05	27.16	GREEN	STABLE GREEN	100.0%
1	0.015	6718.7	1.4850e+05	1.7381e+05	19.88	GREEN	STABLE GREEN	100.0%
1	0.025	8027.7	1.1451e+05	1.3090e+05	13.71	GREEN	INCONCLUSIVE	100.0%
2	0	19349	-2.6383e+06	-4.1399e+07	96.28	GREEN	INCONCLUSIVE	100.0%
2	2.50000e-04	8999.3	-1.5605e+06	-2.8557e+07	69.2	GREEN	STABLE GREEN	100.0%
2	5.00000e-04	5070.3	-9.2103e+05	-1.2180e+07	53.79	GREEN	STABLE GREEN	100.0%
2	0.001	2516.3	-3.8134e+05	-3.7349e+06	37.72	GREEN	STABLE GREEN	100.0%
2	0.002	1527.7	-1.0389e+05	-6.2956e+05	24.99	GREEN	STABLE GREEN	100.0%
2	0.0035	1391.3	-2.408e+04	2.763e+04	18.24	RED	INCONCLUSIVE	100.0%
2	0.0037856	1405.7	-1.548e+04	4.79e+04	17.61	RED	INCONCLUSIVE	100.0%
2	0.005	1452	-4133	2.821e+04	15.18	RED	INCONCLUSIVE	100.0%
2	0.0075	1471.3	1.102e+04	1.653e+04	12	GREEN	STABLE GREEN	100.0%
2	0.01	1534.7	1.888e+04	2.358e+04	10.17	GREEN	STABLE GREEN	100.0%
2	0.015	1635	1.6e+04	1.812e+04	8.121	GREEN	STABLE GREEN	100.0%
2	0.025	1777.3	1.139e+04	1.423e+04	6.253	GREEN	INCONCLUSIVE	100.0%
7	0	19109	-1.7424e+06	-3.8103e+07	76.42	GREEN	INCONCLUSIVE	100.0%
7	2.50000e-04	9583	-7.1711e+05	-2.5157e+07	52.43	GREEN	STABLE GREEN	100.0%
7	5.00000e-04	6530	-4.3568e+05	-9.6411e+06	42.06	GREEN	STABLE GREEN	100.0%
7	0.001	4280.3	-2.2138e+05	-3.4139e+06	31.71	GREEN	STABLE GREEN	100.0%
7	0.002	3037.3	-7.184e+04	-7.8349e+05	22.71	GREEN	STABLE GREEN	100.0%
7	0.0035	2896	-2.135e+04	-1.0123e+05	16.89	GREEN	INCONCLUSIVE	100.0%
7	0.0043002	2812	-6591	-4.541e+04	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.005	2816.7	1439	1.218e+04	13.85	GREEN	INCONCLUSIVE	100.0%
7	0.0075	2896.3	8329	2.633e+04	10.95	GREEN	STABLE GREEN	100.0%
7	0.01	2948.3	1.334e+04	2.187e+04	9.116	GREEN	STABLE GREEN	100.0%
7	0.015	3068.3	1.91e+04	2.391e+04	7.17	GREEN	STABLE GREEN	100.0%
7	0.025	3305.7	1.854e+04	2.373e+04	5.604	GREEN	INCONCLUSIVE	100.0%

on -> ron gSYN

Objective: PSTH SSE. Qualified stable green/red: 30/0; qualified green fraction 100.0%. Inconclusive: 12; invalid CV: 0; raw green/all: 42/42.

on -> ron gSYN: archived local eligibility versus sampled loss slope



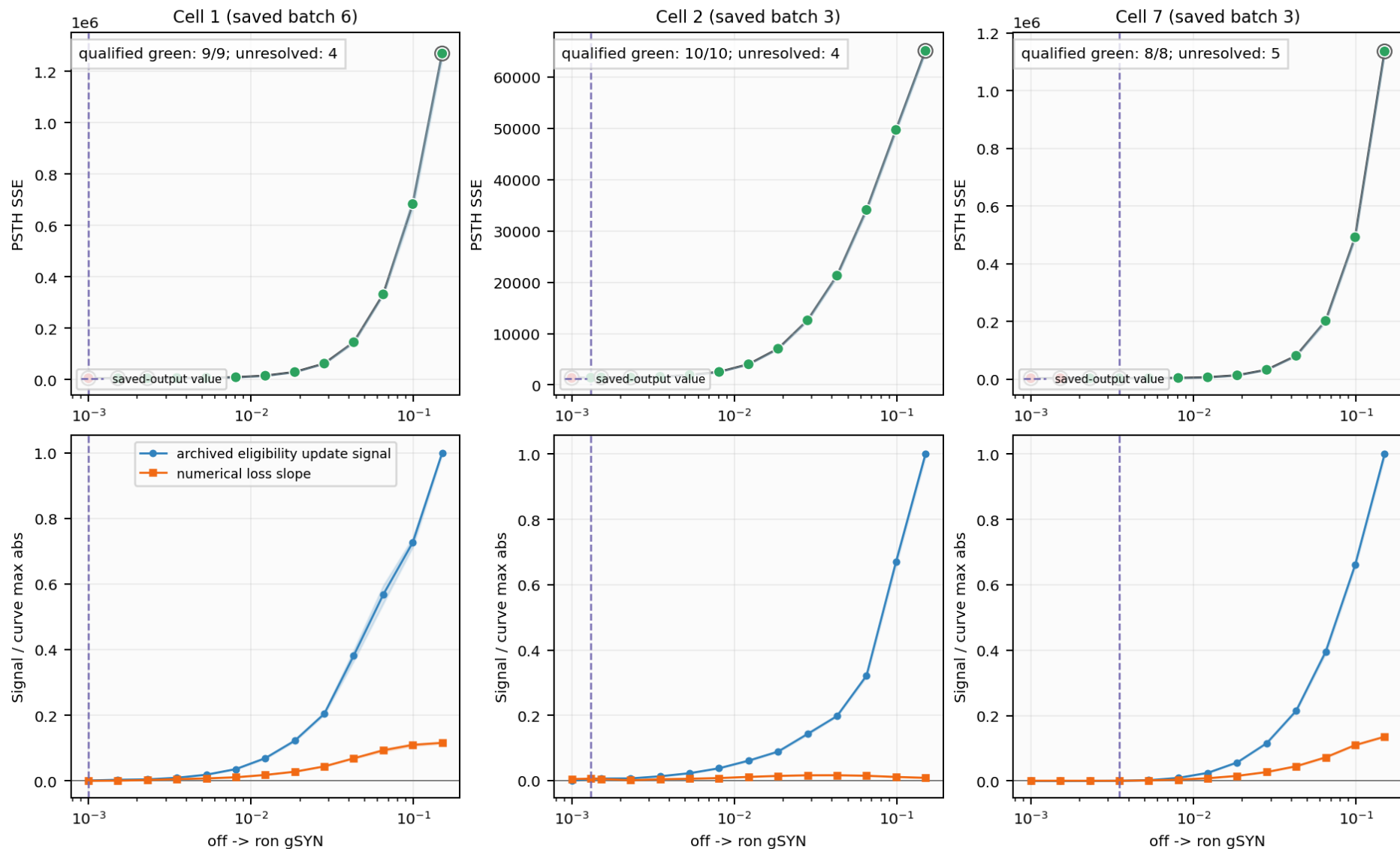
on -> ron gSYN: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.001	10171	-3.5501e+05	-2.6693e+05	6.007	GREEN	INCONCLUSIVE	100.0%
1	0.0015182	10032	-5.5017e+05	-3.6163e+05	6.756	GREEN	STABLE GREEN	100.0%
1	0.0023051	9634.7	-1.0898e+06	-5.0893e+05	8.109	GREEN	STABLE GREEN	100.0%
1	0.0034996	9020.3	-1.4360e+06	-5.1572e+05	10.4	GREEN	STABLE GREEN	100.0%
1	0.0053133	8081	-1.7799e+06	-5.2396e+05	14.21	GREEN	STABLE GREEN	100.0%
1	0.0080669	6613	-1.8072e+06	-4.7080e+05	21.05	GREEN	STABLE GREEN	100.0%
1	0.012247	5040.3	-1.2519e+06	-2.2528e+05	32.42	GREEN	STABLE GREEN	100.0%
1	0.016436	4730	1304	2.506e+04	44.66	GREEN	INCONCLUSIVE	100.0%
1	0.018595	4894.3	7.8118e+05	1.6220e+05	51.09	GREEN	STABLE GREEN	100.0%
1	0.028231	10160	5.2561e+06	9.1660e+05	80.57	GREEN	STABLE GREEN	100.0%
1	0.042862	31793	1.3993e+07	2.0269e+06	126.1	GREEN	STABLE GREEN	100.0%
1	0.065074	95309	2.9680e+07	3.7690e+06	195.6	GREEN	STABLE GREEN	100.0%
1	0.098798	2.68988e+05	5.6584e+07	6.4729e+06	302.2	GREEN	STABLE GREEN	100.0%
1	0.15	7.03250e+05	9.8987e+07	8.4814e+06	465.9	GREEN	INCONCLUSIVE	100.0%
2	0.001	1618.3	-7.443e+04	-6.882e+04	8.836	GREEN	INCONCLUSIVE	100.0%
2	0.0015182	1582.7	-1.0937e+05	-8.035e+04	9.608	GREEN	STABLE GREEN	100.0%
2	0.0023051	1505.7	-1.4331e+05	-8.161e+04	11.05	GREEN	STABLE GREEN	100.0%
2	0.0034996	1437.7	-1.3392e+05	-4.117e+04	13.72	GREEN	STABLE GREEN	100.0%
2	0.0049576	1405.7	2.911e+04	3.337e+04	17.61	GREEN	INCONCLUSIVE	100.0%
2	0.0053133	1422.3	1.1962e+05	5.218e+04	18.52	GREEN	INCONCLUSIVE	100.0%
2	0.0080669	1679.3	8.2405e+05	1.6370e+05	26.99	GREEN	STABLE GREEN	100.0%
2	0.012247	2810.3	2.2012e+06	3.8027e+05	39.98	GREEN	STABLE GREEN	100.0%
2	0.018595	6281.3	4.2034e+06	6.4603e+05	58.09	GREEN	STABLE GREEN	100.0%
2	0.028231	13958	6.6634e+06	8.5308e+05	82.08	GREEN	STABLE GREEN	100.0%
2	0.042862	27693	8.7977e+06	9.5380e+05	110.8	GREEN	STABLE GREEN	100.0%
2	0.065074	49384	1.2538e+07	9.1106e+05	143.1	GREEN	STABLE GREEN	100.0%
2	0.098798	76758	3.3589e+07	7.1162e+05	174.6	GREEN	STABLE GREEN	100.0%
2	0.15	1.05415e+05	7.0152e+07	5.5970e+05	202.1	GREEN	INCONCLUSIVE	100.0%
7	0.001	3810	-3.276e+04	-4.888e+04	6.331	GREEN	INCONCLUSIVE	100.0%
7	0.0015182	3784.7	-4.472e+04	-4.949e+04	6.443	GREEN	INCONCLUSIVE	100.0%
7	0.0023051	3745	-5.609e+04	-7.25e+04	6.655	GREEN	STABLE GREEN	100.0%
7	0.0034996	3618.3	-9.673e+04	-1.1487e+05	7.08	GREEN	STABLE GREEN	100.0%
7	0.0053133	3385.7	-1.5553e+05	-1.2705e+05	8.221	GREEN	STABLE GREEN	100.0%
7	0.0080669	3041	-1.7551e+05	-9.871e+04	10.64	GREEN	STABLE GREEN	100.0%
7	0.011607	2812	5620	7.476e+04	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.012247	2876	5.314e+04	1.0685e+05	15.88	GREEN	INCONCLUSIVE	100.0%
7	0.018595	3986.7	8.4405e+05	3.4500e+05	25.23	GREEN	STABLE GREEN	100.0%
7	0.028231	9798.7	2.7370e+06	8.7023e+05	41.09	GREEN	STABLE GREEN	100.0%
7	0.042862	28464	6.1860e+06	1.6533e+06	66.81	GREEN	STABLE GREEN	100.0%
7	0.065074	77918	1.2280e+07	2.8778e+06	105.8	GREEN	STABLE GREEN	100.0%
7	0.098798	2.08323e+05	2.2347e+07	4.6459e+06	167.1	GREEN	STABLE GREEN	100.0%
7	0.15	5.06767e+05	3.4882e+07	5.8288e+06	257.5	GREEN	INCONCLUSIVE	100.0%

off -> ron gSYN

Objective: PSTH SSE. Qualified stable green/red: 27/0; qualified green fraction 100.0%. Inconclusive: 13; invalid CV: 0; raw green/all: 36/40.

off -> ron gSYN: archived local eligibility versus sampled loss slope



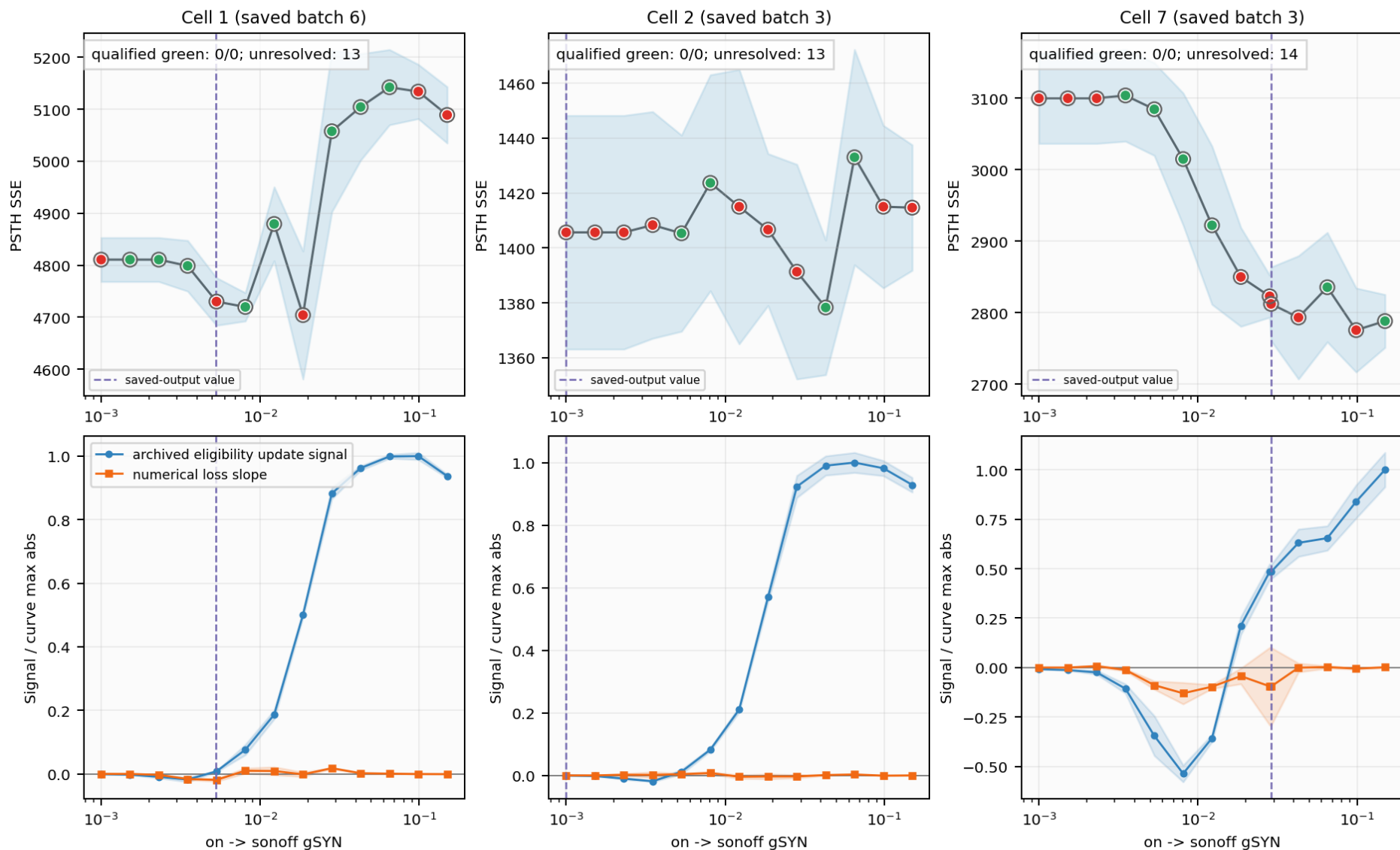
off -> ron gSYN: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.001	4730	5.745e+04	-2.38e+04	44.66	RED	INCONCLUSIVE	100.0%
1	0.0015182	4717.7	2.7769e+05	2.182e+04	45.5	GREEN	INCONCLUSIVE	100.0%
1	0.0023051	4789.3	4.0248e+05	2.0362e+05	46.9	GREEN	INCONCLUSIVE	100.0%
1	0.0034996	5236.7	8.7576e+05	4.6778e+05	48.94	GREEN	STABLE GREEN	100.0%
1	0.0053133	6342	1.8101e+06	6.8499e+05	52.4	GREEN	STABLE GREEN	100.0%
1	0.0080669	8544	3.5074e+06	1.0398e+06	58.02	GREEN	STABLE GREEN	100.0%
1	0.012247	14415	6.8112e+06	1.7438e+06	67.44	GREEN	STABLE GREEN	100.0%
1	0.018595	28755	1.2115e+07	2.7196e+06	83.43	GREEN	STABLE GREEN	100.0%
1	0.028231	61696	2.0252e+07	4.3189e+06	110.2	GREEN	STABLE GREEN	100.0%
1	0.042862	1.44888e+05	3.7862e+07	6.7626e+06	154	GREEN	STABLE GREEN	100.0%
1	0.065074	3.31405e+05	5.6320e+07	9.2052e+06	222.8	GREEN	STABLE GREEN	100.0%
1	0.098798	6.83227e+05	7.2056e+07	1.0839e+07	328.5	GREEN	STABLE GREEN	100.0%
1	0.15	1.26975e+06	9.9214e+07	1.1455e+07	490.6	GREEN	INCONCLUSIVE	100.0%
2	0.001	1354.7	-3.68e+04	1.6461e+05	16.82	RED	INCONCLUSIVE	100.0%
2	0.0013098	1405.7	2.1388e+05	2.0293e+05	17.61	GREEN	STABLE GREEN	100.0%
2	0.0015182	1453.3	2.2474e+05	1.7646e+05	17.86	GREEN	INCONCLUSIVE	100.0%
2	0.0023051	1437	2.3350e+05	3.358e+04	19.28	GREEN	INCONCLUSIVE	100.0%
2	0.0034996	1575.7	4.7150e+05	1.4362e+05	21.88	GREEN	STABLE GREEN	100.0%
2	0.0053133	1912	8.1370e+05	2.0030e+05	25.8	GREEN	STABLE GREEN	100.0%
2	0.0080669	2525.7	1.3625e+06	2.7615e+05	32.24	GREEN	STABLE GREEN	100.0%
2	0.012247	4018.3	2.2053e+06	4.0446e+05	41.24	GREEN	STABLE GREEN	100.0%
2	0.018595	7042.3	3.2103e+06	5.1532e+05	53.8	GREEN	STABLE GREEN	100.0%
2	0.028231	12577	5.1696e+06	5.8263e+05	70.13	GREEN	STABLE GREEN	100.0%
2	0.042862	21285	7.1609e+06	5.8774e+05	89.28	GREEN	STABLE GREEN	100.0%
2	0.065074	34088	1.1642e+07	5.3112e+05	110.8	GREEN	STABLE GREEN	100.0%
2	0.098798	49682	2.4259e+07	3.9822e+05	131.9	GREEN	STABLE GREEN	100.0%
2	0.15	65083	3.6223e+07	3.0078e+05	149.5	GREEN	INCONCLUSIVE	100.0%
7	0.001	2736.7	-8377	5.339e+04	11.63	RED	INCONCLUSIVE	100.0%
7	0.0015182	2764.3	-1.17e+04	5.422e+04	12.25	RED	INCONCLUSIVE	100.0%
7	0.0023051	2808	9874	3.479e+04	13.33	GREEN	INCONCLUSIVE	100.0%
7	0.0034996	2812	4.021e+04	4.048e+04	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.0053133	2987.7	1.9932e+05	1.5247e+05	18.31	GREEN	STABLE GREEN	100.0%
7	0.0080669	3640	8.5537e+05	3.6333e+05	24.52	GREEN	STABLE GREEN	100.0%
7	0.012247	5961.3	2.2869e+06	7.5914e+05	35.73	GREEN	STABLE GREEN	100.0%
7	0.018595	12744	5.2563e+06	1.4345e+06	54.58	GREEN	STABLE GREEN	100.0%
7	0.028231	31919	1.0729e+07	2.5191e+06	85.86	GREEN	STABLE GREEN	100.0%
7	0.042862	80533	1.9964e+07	4.1671e+06	135.3	GREEN	STABLE GREEN	100.0%
7	0.065074	2.01571e+05	3.6757e+07	6.7018e+06	211.2	GREEN	STABLE GREEN	100.0%
7	0.098798	4.91727e+05	6.1479e+07	1.0185e+07	328.7	GREEN	STABLE GREEN	100.0%
7	0.15	1.13608e+06	9.3045e+07	1.2585e+07	507	GREEN	INCONCLUSIVE	100.0%

on -> sonoff gSYN

Objective: PSTH SSE. Qualified stable green/red: 0/0; qualified green fraction N/A. Inconclusive: 40; invalid CV: 0; raw green/all: 18/40.

on -> sonoff gSYN: archived local eligibility versus sampled loss slope



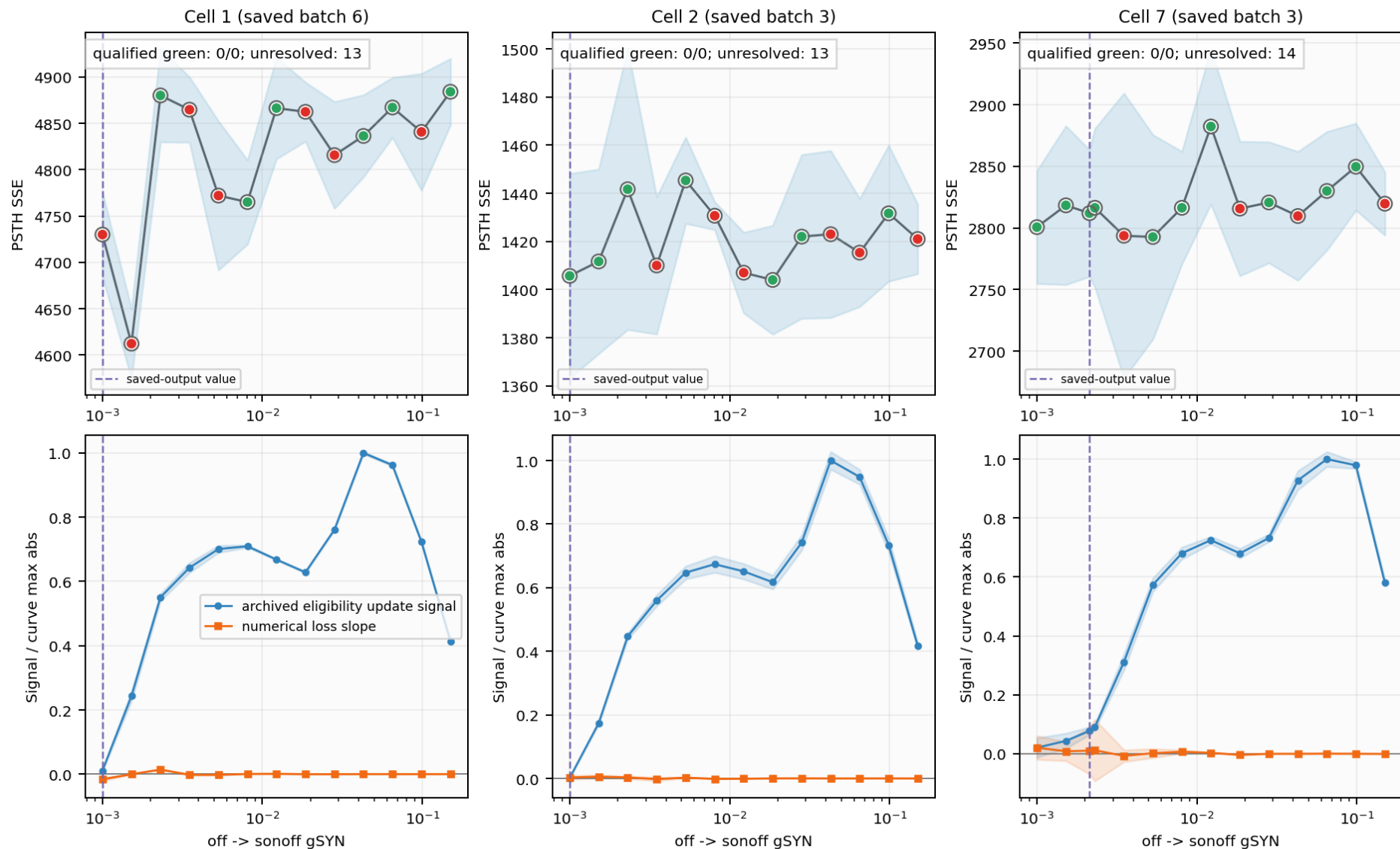
on -> sonoff gSYN: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.001	4810.7	-1146	0	44.73	RED	INCONCLUSIVE	100.0%
1	0.0015182	4810.7	-3092	-1.5522e-10	44.73	GREEN	INCONCLUSIVE	100.0%
1	0.0023051	4810.7	-1.193e+04	-3989	44.73	GREEN	INCONCLUSIVE	100.0%
1	0.0034996	4798.7	-2.307e+04	-2.109e+04	44.73	GREEN	INCONCLUSIVE	100.0%
1	0.0053133	4730	1.049e+04	-2.432e+04	44.66	RED	INCONCLUSIVE	100.0%
1	0.0080669	4719.7	9.906e+04	1.29e+04	44.43	GREEN	INCONCLUSIVE	100.0%
1	0.012247	4879.3	2.4264e+05	1.206e+04	43.45	GREEN	INCONCLUSIVE	100.0%
1	0.018595	4704	6.5198e+05	-2080	40.88	RED	INCONCLUSIVE	100.0%
1	0.028231	5057.7	1.1489e+06	2.339e+04	37.87	GREEN	INCONCLUSIVE	100.0%
1	0.042862	5104.3	1.2527e+06	2602	36.37	GREEN	INCONCLUSIVE	100.0%
1	0.065074	5142.3	1.3002e+06	933.3	35.66	GREEN	INCONCLUSIVE	100.0%
1	0.098798	5134	1.3017e+06	-498	35.38	RED	INCONCLUSIVE	100.0%
1	0.15	5089	1.2201e+06	-878.9	35.12	RED	INCONCLUSIVE	100.0%
2	0.001	1405.7	-194.8	0	17.61	RED	INCONCLUSIVE	100.0%
2	0.0015182	1405.7	-791.7	3.8805e-11	17.61	RED	INCONCLUSIVE	100.0%
2	0.0023051	1405.7	-4307	886.5	17.61	RED	INCONCLUSIVE	100.0%
2	0.0034996	1408.3	-7848	689	17.58	RED	INCONCLUSIVE	100.0%
2	0.0053133	1405.3	4890	1647	17.53	GREEN	INCONCLUSIVE	100.0%
2	0.0080669	1423.7	3.461e+04	3191	17.45	GREEN	INCONCLUSIVE	100.0%
2	0.012247	1415	8.844e+04	-1771	17.09	RED	INCONCLUSIVE	100.0%
2	0.018595	1406.7	2.3923e+05	-1423	15.29	RED	INCONCLUSIVE	100.0%
2	0.028231	1391.3	3.8764e+05	-1312	12.78	RED	INCONCLUSIVE	100.0%
2	0.042862	1378.3	4.1575e+05	441.6	12.06	GREEN	INCONCLUSIVE	100.0%
2	0.065074	1433	4.1990e+05	1272	11.83	GREEN	INCONCLUSIVE	100.0%
2	0.098798	1415	4.1229e+05	-324.4	11.77	RED	INCONCLUSIVE	100.0%
2	0.15	1414.7	3.8998e+05	-6.51	11.62	RED	INCONCLUSIVE	100.0%
7	0.001	3099.7	-1502	0	18.29	RED	INCONCLUSIVE	100.0%
7	0.0015182	3099.7	-2346	0	18.29	RED	INCONCLUSIVE	100.0%
7	0.0023051	3099.7	-4508	1330	18.29	RED	INCONCLUSIVE	100.0%
7	0.0034996	3103.7	-1.975e+04	-2214	18.27	GREEN	INCONCLUSIVE	100.0%
7	0.0053133	3084.3	-6.375e+04	-1.647e+04	18.22	GREEN	INCONCLUSIVE	100.0%
7	0.0080669	3014.7	-9.909e+04	-2.406e+04	17.87	GREEN	INCONCLUSIVE	100.0%
7	0.012247	2922	-6.597e+04	-1.789e+04	17.2	GREEN	INCONCLUSIVE	100.0%
7	0.018595	2849.7	3.91e+04	-7983	16.05	RED	INCONCLUSIVE	100.0%
7	0.028231	2822.7	8.953e+04	-1.752e+04	15.06	RED	INCONCLUSIVE	100.0%
7	0.02881	2812	8.939e+04	-1.773e+04	15.06	RED	INCONCLUSIVE	100.0%
7	0.042862	2793	1.1652e+05	-89.82	14.51	RED	INCONCLUSIVE	100.0%
7	0.065074	2835.3	1.2095e+05	442.5	14.25	GREEN	INCONCLUSIVE	100.0%
7	0.098798	2775.3	1.5511e+05	-974.4	13.96	RED	INCONCLUSIVE	100.0%
7	0.15	2788	1.8488e+05	247.4	13.96	GREEN	INCONCLUSIVE	100.0%

off -> sonoff gSYN

Objective: PSTH SSE. Qualified stable green/red: 0/0; qualified green fraction N/A. Inconclusive: 40; invalid CV: 0; raw green/all: 23/40.

off -> sonoff gSYN: archived local eligibility versus sampled loss slope



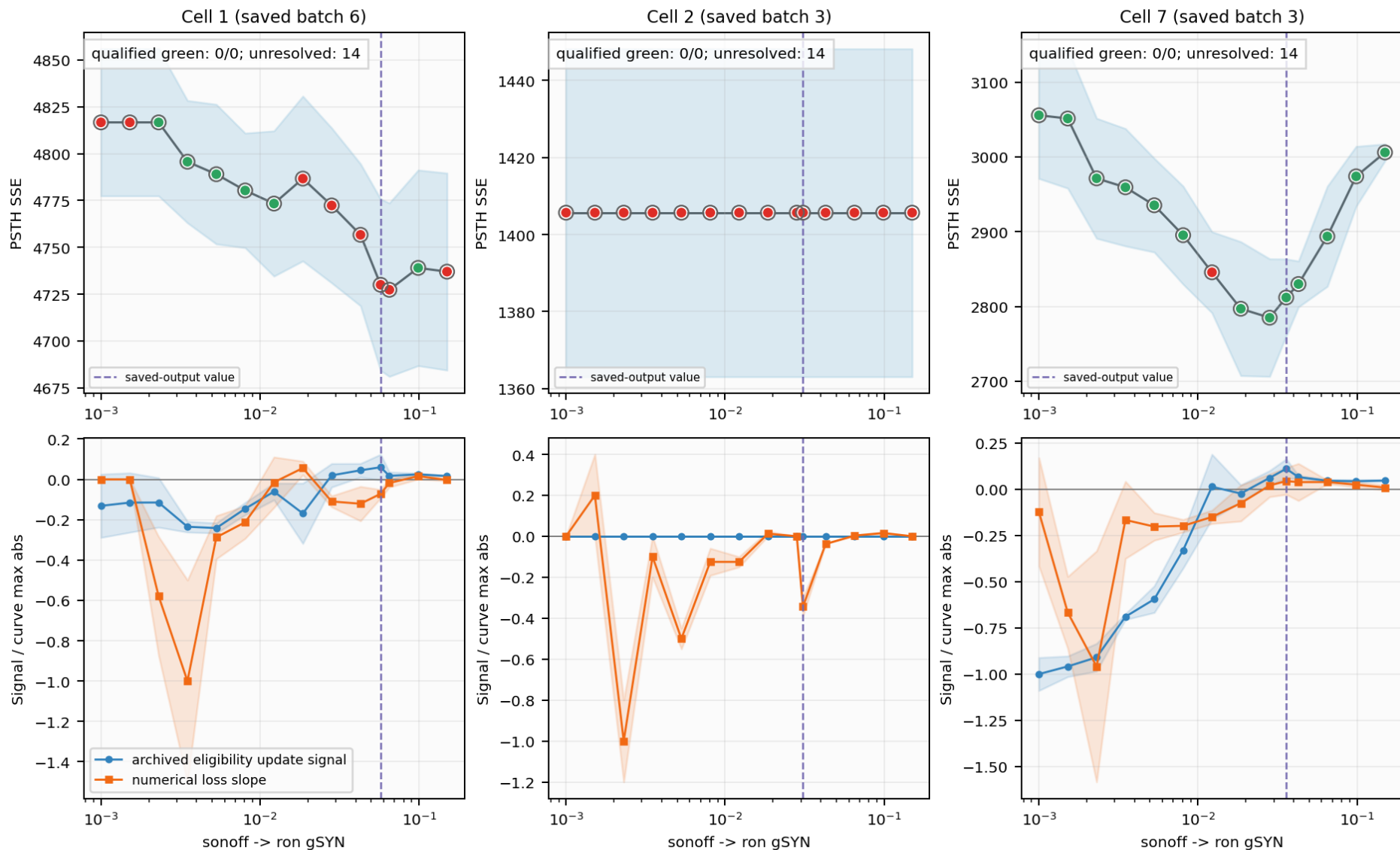
off -> sonoff gSYN: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.001	4730	1.5587e+05	-2.2705e+05	44.66	RED	INCONCLUSIVE	100.0%
1	0.0015182	4612.3	3.4451e+06	-1798	43.85	RED	INCONCLUSIVE	100.0%
1	0.0023051	4880	7.8024e+06	2.0000e+05	41.35	GREEN	INCONCLUSIVE	100.0%
1	0.0034996	4864.7	9.1106e+06	-2.803e+04	39.7	RED	INCONCLUSIVE	100.0%
1	0.0053133	4772	9.9341e+06	-3.181e+04	38.24	RED	INCONCLUSIVE	100.0%
1	0.0080669	4765	1.0057e+07	8093	37.56	GREEN	INCONCLUSIVE	100.0%
1	0.012247	4866.3	9.4686e+06	1.436e+04	37.09	GREEN	INCONCLUSIVE	100.0%
1	0.018595	4862.3	8.9069e+06	-2303	36.8	RED	INCONCLUSIVE	100.0%
1	0.028231	4815.7	1.0789e+07	-2359	36.67	RED	INCONCLUSIVE	100.0%
1	0.042862	4836.3	1.4175e+07	1400	36.49	GREEN	INCONCLUSIVE	100.0%
1	0.065074	4867	1.3637e+07	522.3	36.31	GREEN	INCONCLUSIVE	100.0%
1	0.098798	4840.7	1.0254e+07	-134.7	36.3	RED	INCONCLUSIVE	100.0%
1	0.15	4884	5.8609e+06	846.3	36.23	GREEN	INCONCLUSIVE	100.0%
2	0.001	1405.7	4994	1.158e+04	17.61	GREEN	INCONCLUSIVE	100.0%
2	0.0015182	1411.7	6.9214e+05	2.212e+04	17.52	GREEN	INCONCLUSIVE	100.0%
2	0.0023051	1441.7	1.7796e+06	1.246e+04	16.67	GREEN	INCONCLUSIVE	100.0%
2	0.0034996	1410	2.2284e+06	-8246	15.36	RED	INCONCLUSIVE	100.0%
2	0.0053133	1445.3	2.5786e+06	9630	14.37	GREEN	INCONCLUSIVE	100.0%
2	0.0080669	1430.7	2.6841e+06	-5459	13.84	RED	INCONCLUSIVE	100.0%
2	0.012247	1407	2.5939e+06	-3601	13.33	RED	INCONCLUSIVE	100.0%
2	0.018595	1404	2.4584e+06	456.8	12.78	GREEN	INCONCLUSIVE	100.0%
2	0.028231	1422	2.9546e+06	1153	12.74	GREEN	INCONCLUSIVE	100.0%
2	0.042862	1423	3.9830e+06	-95.85	12.55	RED	INCONCLUSIVE	100.0%
2	0.065074	1415.3	3.7771e+06	-15.76	12.49	RED	INCONCLUSIVE	100.0%
2	0.098798	1431.7	2.9243e+06	209.3	12.44	GREEN	INCONCLUSIVE	100.0%
2	0.15	1421	1.6641e+06	-208.3	12.37	RED	INCONCLUSIVE	100.0%
7	0.001	2800.7	3.396e+04	3.409e+04	15.3	GREEN	INCONCLUSIVE	100.0%
7	0.0015182	2818.3	7.19e+04	1.379e+04	15.25	GREEN	INCONCLUSIVE	100.0%
7	0.002133	2812	1.2765e+05	1.743e+04	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.0023051	2816.3	1.5007e+05	1.963e+04	15	GREEN	INCONCLUSIVE	100.0%
7	0.0034996	2793.7	5.1087e+05	-1.166e+04	14.22	RED	INCONCLUSIVE	100.0%
7	0.0053133	2792.7	9.4171e+05	3081	13.56	GREEN	INCONCLUSIVE	100.0%
7	0.0080669	2816.3	1.1177e+06	1.145e+04	13.21	GREEN	INCONCLUSIVE	100.0%
7	0.012247	2882.3	1.1905e+06	5347	12.99	GREEN	INCONCLUSIVE	100.0%
7	0.018595	2815.7	1.1170e+06	-6126	12.83	RED	INCONCLUSIVE	100.0%
7	0.028231	2820.7	1.2028e+06	14.26	12.82	GREEN	INCONCLUSIVE	100.0%
7	0.042862	2809.7	1.5240e+06	-89.79	12.87	RED	INCONCLUSIVE	100.0%
7	0.065074	2830	1.6419e+06	783.5	12.93	GREEN	INCONCLUSIVE	100.0%
7	0.098798	2849.7	1.6078e+06	118.9	12.92	GREEN	INCONCLUSIVE	100.0%
7	0.15	2819.7	9.5346e+05	-585.9	13.01	RED	INCONCLUSIVE	100.0%

sonoff -> ron gSYN

Objective: PSTH SSE. Qualified stable green/red: 0/0; qualified green fraction N/A. Inconclusive: 42; invalid CV: 0; raw green/all: 19/42.

sonoff -> ron gSYN: archived local eligibility versus sampled loss slope



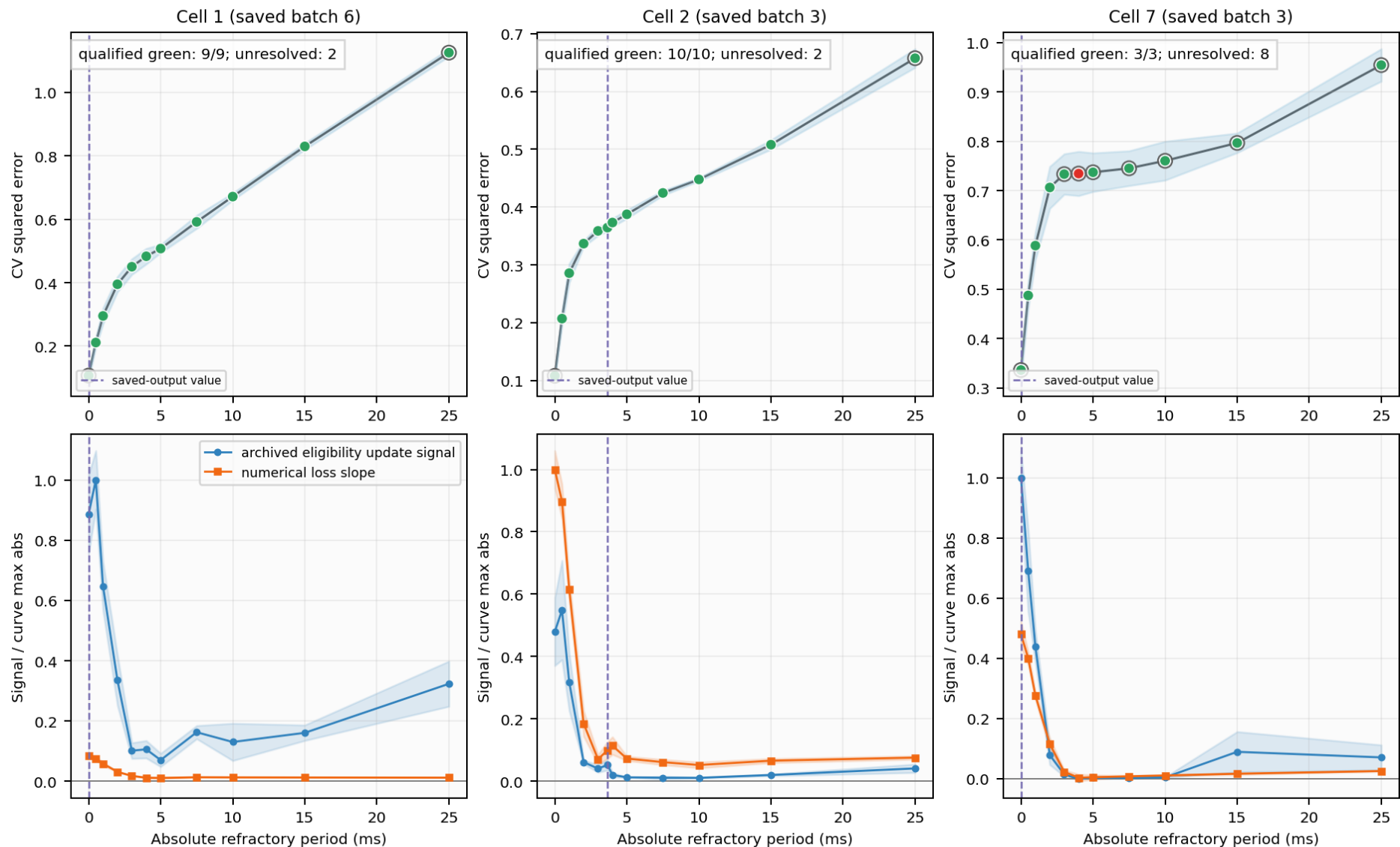
sonoff -> ron gSYN: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.001	4816.7	-1585	0	44.73	RED	INCONCLUSIVE	100.0%
1	0.0015182	4816.7	-1390	1.5522e-10	44.73	RED	INCONCLUSIVE	100.0%
1	0.0023051	4816.7	-1386	-6981	44.73	GREEN	INCONCLUSIVE	100.0%
1	0.0034996	4795.7	-2834	-1.206e+04	44.72	GREEN	INCONCLUSIVE	100.0%
1	0.0053133	4789	-2911	-3466	44.72	GREEN	INCONCLUSIVE	100.0%
1	0.0080669	4780.3	-1756	-2562	44.7	GREEN	INCONCLUSIVE	100.0%
1	0.012247	4773.3	-731.2	-175.3	44.69	GREEN	INCONCLUSIVE	100.0%
1	0.018595	4786.7	-2037	675.8	44.69	RED	INCONCLUSIVE	100.0%
1	0.028231	4772.3	238.6	-1322	44.67	RED	INCONCLUSIVE	100.0%
1	0.042862	4756.7	540.7	-1458	44.66	RED	INCONCLUSIVE	100.0%
1	0.057351	4730	723.9	-865.2	44.66	RED	INCONCLUSIVE	100.0%
1	0.065074	4727.3	210.2	-216.5	44.66	RED	INCONCLUSIVE	100.0%
1	0.098798	4739	301.4	193.1	44.65	GREEN	INCONCLUSIVE	100.0%
1	0.15	4737	198.9	-39.06	44.65	RED	INCONCLUSIVE	100.0%
2	0.001	1405.7	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	0.0015182	1405.7	0	3.8805e-11	17.61	RED	INCONCLUSIVE	100.0%
2	0.0023051	1405.7	0	-1.9403e-10	17.61	RED	INCONCLUSIVE	100.0%
2	0.0034996	1405.7	0	-1.9403e-11	17.61	RED	INCONCLUSIVE	100.0%
2	0.0053133	1405.7	0	-9.7013e-11	17.61	RED	INCONCLUSIVE	100.0%
2	0.0080669	1405.7	0	-2.4253e-11	17.61	RED	INCONCLUSIVE	100.0%
2	0.012247	1405.7	0	-2.4253e-11	17.61	RED	INCONCLUSIVE	100.0%
2	0.018595	1405.7	0	2.4253e-12	17.61	RED	INCONCLUSIVE	100.0%
2	0.028231	1405.7	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	0.030888	1405.7	0	-6.6696e-11	17.61	RED	INCONCLUSIVE	100.0%
2	0.042862	1405.7	0	-7.2760e-12	17.61	RED	INCONCLUSIVE	100.0%
2	0.065074	1405.7	0	6.0633e-13	17.61	RED	INCONCLUSIVE	100.0%
2	0.098798	1405.7	0	3.0316e-12	17.61	RED	INCONCLUSIVE	100.0%
2	0.15	1405.7	0	0	17.61	RED	INCONCLUSIVE	100.0%
7	0.001	3055.7	-6.809e+04	-8362	18.06	GREEN	INCONCLUSIVE	100.0%
7	0.0015182	3051.3	-6.527e+04	-4.542e+04	17.99	GREEN	INCONCLUSIVE	100.0%
7	0.0023051	2971.3	-6.192e+04	-6.529e+04	17.81	GREEN	INCONCLUSIVE	100.0%
7	0.0034996	2959.3	-4.695e+04	-1.131e+04	17.7	GREEN	INCONCLUSIVE	100.0%
7	0.0053133	2935.3	-4.051e+04	-1.375e+04	17.47	GREEN	INCONCLUSIVE	100.0%
7	0.0080669	2895.3	-2.244e+04	-1.348e+04	17.09	GREEN	INCONCLUSIVE	100.0%
7	0.012247	2845.7	933.8	-1.021e+04	16.57	RED	INCONCLUSIVE	100.0%
7	0.018595	2797	-1559	-5117	16.1	GREEN	INCONCLUSIVE	100.0%
7	0.028231	2785	4214	1415	15.54	GREEN	INCONCLUSIVE	100.0%
7	0.035884	2812	7563	3032	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.042862	2830	4508	2648	14.79	GREEN	INCONCLUSIVE	100.0%
7	0.065074	2893.7	3138	2674	14.04	GREEN	INCONCLUSIVE	100.0%
7	0.098798	2974	2994	1684	13.4	GREEN	INCONCLUSIVE	100.0%
7	0.15	3006	3222	625	12.8	GREEN	INCONCLUSIVE	100.0%

Absolute refractory period (ms)

Objective: CV squared error. Qualified stable green/red: 22/0; qualified green fraction 100.0%. Inconclusive: 12; invalid CV: 0; raw green/all: 33/34.

Absolute refractory period (ms): archived local eligibility versus sampled loss slope



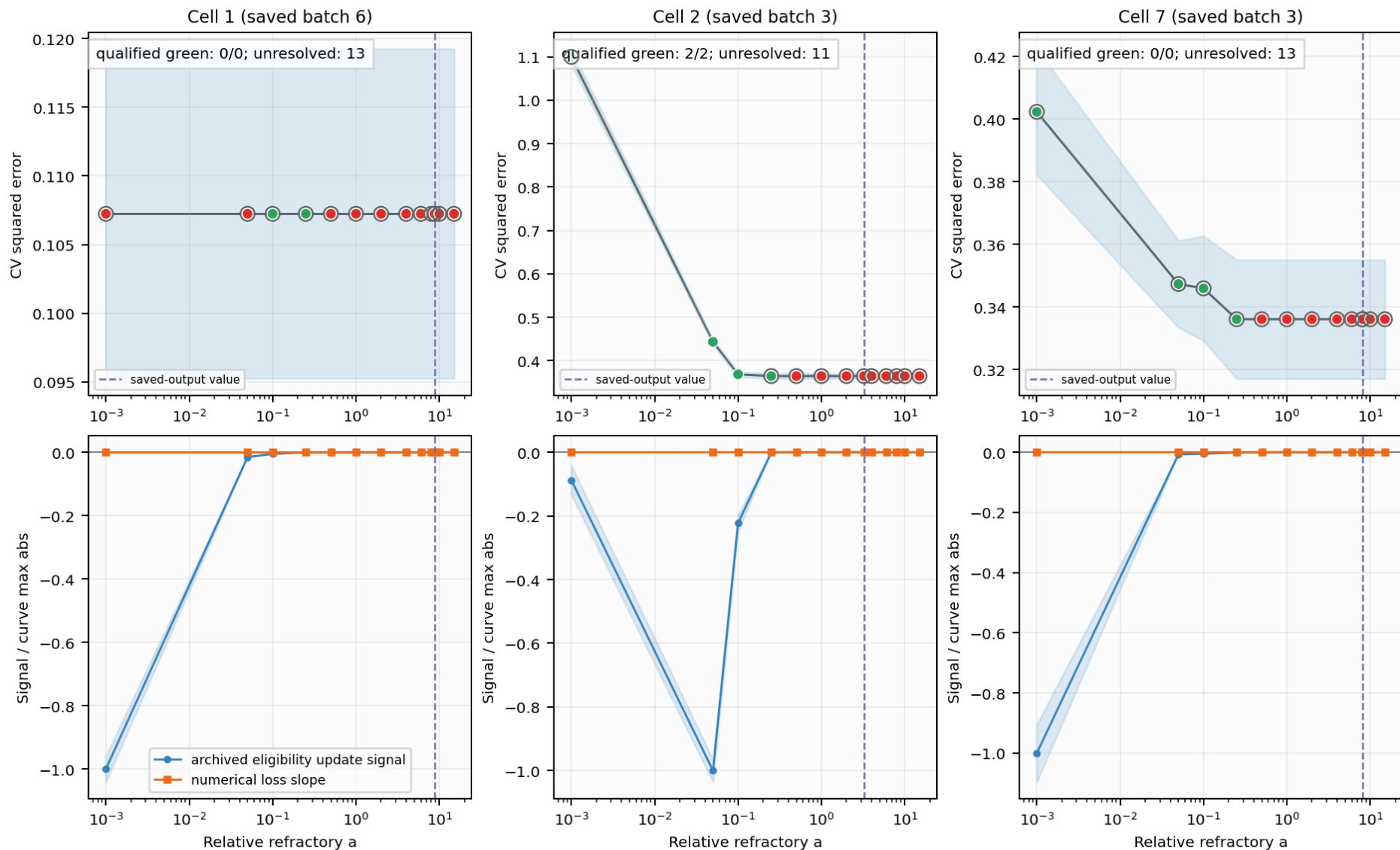
Absolute refractory period (ms): numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0	0.10724	2.213	0.2082	44.66	GREEN	INCONCLUSIVE	0.0%
1	0.5	0.21133	2.498	0.188	42.55	GREEN	STABLE GREEN	100.0%
1	1	0.29525	1.617	0.1451	41.15	GREEN	STABLE GREEN	100.0%
1	2	0.39496	0.8371	0.0775	39.82	GREEN	STABLE GREEN	100.0%
1	3	0.45025	0.2531	0.04414	39.09	GREEN	STABLE GREEN	100.0%
1	4	0.48324	0.2642	0.02861	38.57	GREEN	STABLE GREEN	100.0%
1	5	0.50747	0.1752	0.02693	38.19	GREEN	STABLE GREEN	100.0%
1	7.5	0.59167	0.4061	0.03276	37.41	GREEN	STABLE GREEN	100.0%
1	10	0.67129	0.3251	0.03178	36.5	GREEN	STABLE GREEN	100.0%
1	15	0.82957	0.4012	0.03095	34.1	GREEN	STABLE GREEN	100.0%
1	25	1.125	0.8075	0.02954	28.37	GREEN	INCONCLUSIVE	0.0%
2	0	0.10804	0.09488	0.1979	19.75	GREEN	INCONCLUSIVE	0.0%
2	0.5	0.207	0.1086	0.1776	18.74	GREEN	STABLE GREEN	100.0%
2	1	0.28566	0.06299	0.1219	17.92	GREEN	STABLE GREEN	100.0%
2	2	0.33676	0.0121	0.03657	17.55	GREEN	STABLE GREEN	100.0%
2	3	0.3588	0.007998	0.01396	17.65	GREEN	STABLE GREEN	100.0%
2	3.6539	0.36448	0.01031	0.01963	17.61	GREEN	STABLE GREEN	100.0%
2	4	0.37328	0.003997	0.02254	17.56	GREEN	STABLE GREEN	100.0%
2	5	0.38748	0.002462	0.01437	17.48	GREEN	STABLE GREEN	100.0%
2	7.5	0.42444	0.002219	0.01208	17.18	GREEN	STABLE GREEN	100.0%
2	10	0.44787	0.002123	0.01026	16.82	GREEN	STABLE GREEN	100.0%
2	15	0.50796	0.003938	0.013	16.45	GREEN	STABLE GREEN	100.0%
2	25	0.65765	0.008118	0.01497	15.68	GREEN	INCONCLUSIVE	0.0%
7	0	0.33612	0.6303	0.3025	15.06	GREEN	INCONCLUSIVE	0.0%
7	0.5	0.48735	0.4354	0.2523	13.89	GREEN	STABLE GREEN	100.0%
7	1	0.58846	0.277	0.1741	13.27	GREEN	STABLE GREEN	100.0%
7	2	0.70637	0.04918	0.0725	12.73	GREEN	STABLE GREEN	100.0%
7	3	0.73345	0.008585	0.01408	12.68	GREEN	INCONCLUSIVE	100.0%
7	4	0.73453	-5.6482e-04	0.001858	12.66	RED	INCONCLUSIVE	100.0%
7	5	0.73717	0.002377	0.002814	12.65	GREEN	INCONCLUSIVE	100.0%
7	7.5	0.74529	0.001324	0.004666	12.63	GREEN	INCONCLUSIVE	100.0%
7	10	0.7605	0.002452	0.006471	12.54	GREEN	INCONCLUSIVE	100.0%
7	15	0.79674	0.05601	0.0101	12.35	GREEN	INCONCLUSIVE	100.0%
7	25	0.95466	0.04434	0.01579	11.57	GREEN	INCONCLUSIVE	0.0%

Relative refractory a

Objective: CV squared error. Qualified stable green/red: 2/0; qualified green fraction 100.0%. Inconclusive: 37; invalid CV: 0; raw green/all: 10/39.

Relative refractory a: archived local eligibility versus sampled loss slope



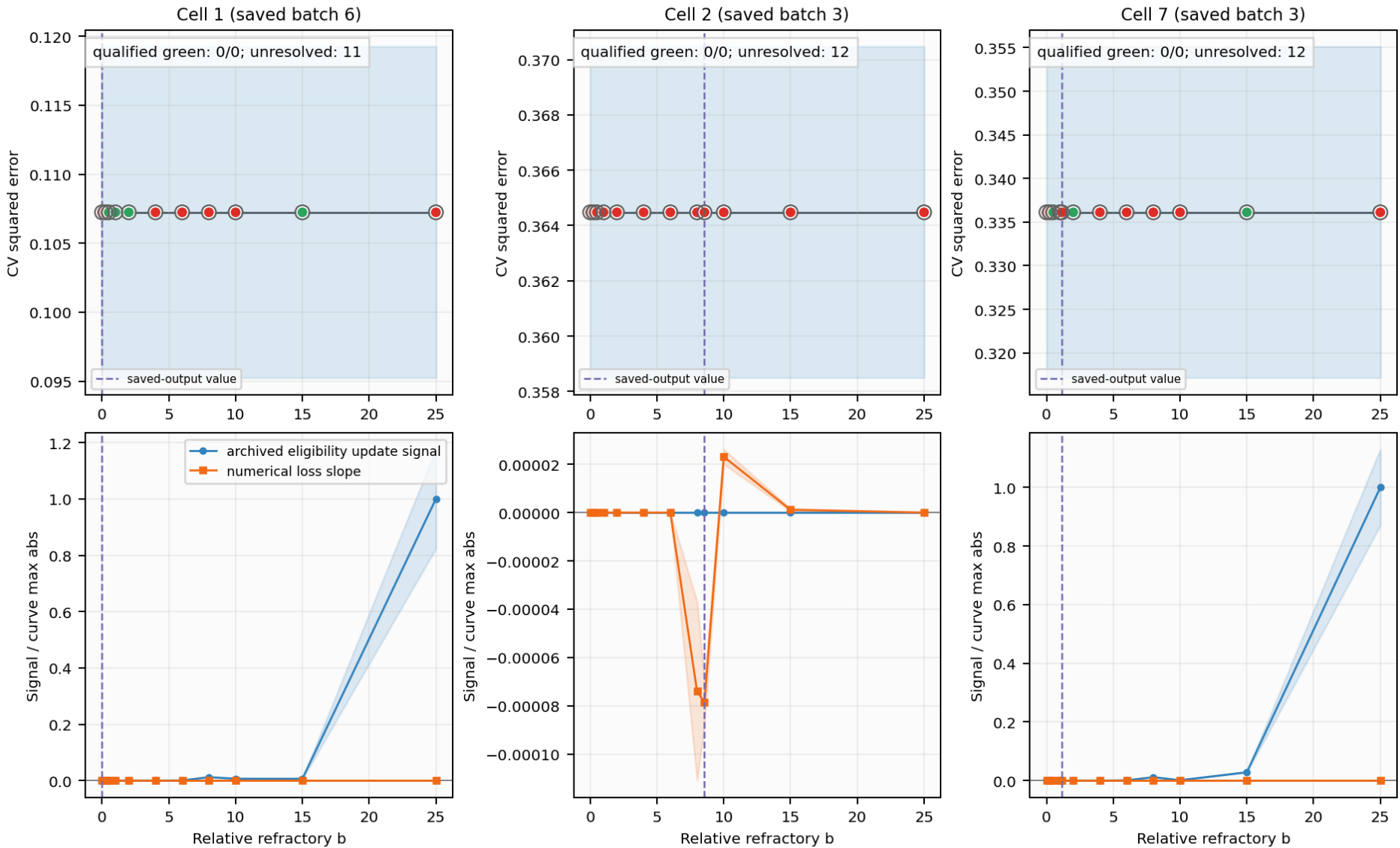
Relative refractory a: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.001	0.10724	-9.0903e+05	0	44.66	RED	INCONCLUSIVE	0.0%
1	0.05	0.10724	-1.399e+04	0	44.66	RED	INCONCLUSIVE	100.0%
1	0.1	0.10724	-4826	-1.8504e-17	44.66	GREEN	INCONCLUSIVE	100.0%
1	0.25	0.10724	-426.9	-9.2519e-18	44.66	GREEN	INCONCLUSIVE	100.0%
1	0.5	0.10724	-22.07	1.1565e-17	44.66	RED	INCONCLUSIVE	100.0%
1	1	0.10724	-0.2928	5.7824e-18	44.66	RED	INCONCLUSIVE	100.0%
1	2	0.10724	-0.001923	2.8912e-18	44.66	RED	INCONCLUSIVE	100.0%
1	4	0.10724	-2.1103e-05	0	44.66	RED	INCONCLUSIVE	100.0%
1	6	0.10724	-3.8296e-07	0	44.66	RED	INCONCLUSIVE	100.0%
1	8	0.10724	-7.4361e-09	4.6259e-18	44.66	RED	INCONCLUSIVE	100.0%
1	8.796	0.10724	-1.8590e-09	3.4694e-18	44.66	RED	INCONCLUSIVE	100.0%
1	10	0.10724	0	5.4933e-18	44.66	RED	INCONCLUSIVE	100.0%
1	15	0.10724	0	0	44.66	RED	INCONCLUSIVE	0.0%
2	0.001	1.0999	-4135	-13.4	1.957	GREEN	INCONCLUSIVE	0.0%
2	0.05	0.44332	-4.703e+04	-7.509	17.17	GREEN	STABLE GREEN	100.0%
2	0.1	0.3684	-1.048e+04	-1.13	17.59	GREEN	STABLE GREEN	100.0%
2	0.25	0.36448	-18.52	-0.01635	17.61	GREEN	INCONCLUSIVE	100.0%
2	0.5	0.36448	-6.0173e-10	0	17.61	RED	INCONCLUSIVE	100.0%
2	1	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	2	0.36448	0	3.7007e-17	17.61	RED	INCONCLUSIVE	100.0%
2	3.245	0.36448	0	-1.8504e-17	17.61	RED	INCONCLUSIVE	100.0%
2	4	0.36448	0	-3.7007e-17	17.61	RED	INCONCLUSIVE	100.0%
2	6	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	8	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	10	0.36448	0	-4.6259e-18	17.61	RED	INCONCLUSIVE	100.0%
2	15	0.36448	0	0	17.61	RED	INCONCLUSIVE	0.0%
7	0.001	0.40235	-1.6025e+06	-1.122	14.54	GREEN	INCONCLUSIVE	0.0%
7	0.05	0.34737	-1.156e+04	-0.5805	14.93	GREEN	INCONCLUSIVE	100.0%
7	0.1	0.34597	-7989	-0.03738	14.99	GREEN	INCONCLUSIVE	100.0%
7	0.25	0.33612	-1238	-0.04107	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.5	0.33612	-73.94	9.2519e-18	15.06	RED	INCONCLUSIVE	100.0%
7	1	0.33612	-0.8488	4.6259e-18	15.06	RED	INCONCLUSIVE	100.0%
7	2	0.33612	-0.005826	2.3130e-18	15.06	RED	INCONCLUSIVE	100.0%
7	4	0.33612	-8.9906e-05	0	15.06	RED	INCONCLUSIVE	100.0%
7	6	0.33612	-1.6473e-06	0	15.06	RED	INCONCLUSIVE	100.0%
7	8	0.33612	-3.0025e-08	2.9606e-16	15.06	RED	INCONCLUSIVE	100.0%
7	8.1406	0.33612	-2.2721e-08	3.9031e-16	15.06	RED	INCONCLUSIVE	100.0%
7	10	0.33612	-8.1148e-10	1.2721e-17	15.06	RED	INCONCLUSIVE	100.0%
7	15	0.33612	0	0	15.06	RED	INCONCLUSIVE	0.0%

Relative refractory b

Objective: CV squared error. Qualified stable green/red: 0/0; qualified green fraction N/A. Inconclusive: 35; invalid CV: 0; raw green/all: 7/35.

Relative refractory b: archived local eligibility versus sampled loss slope



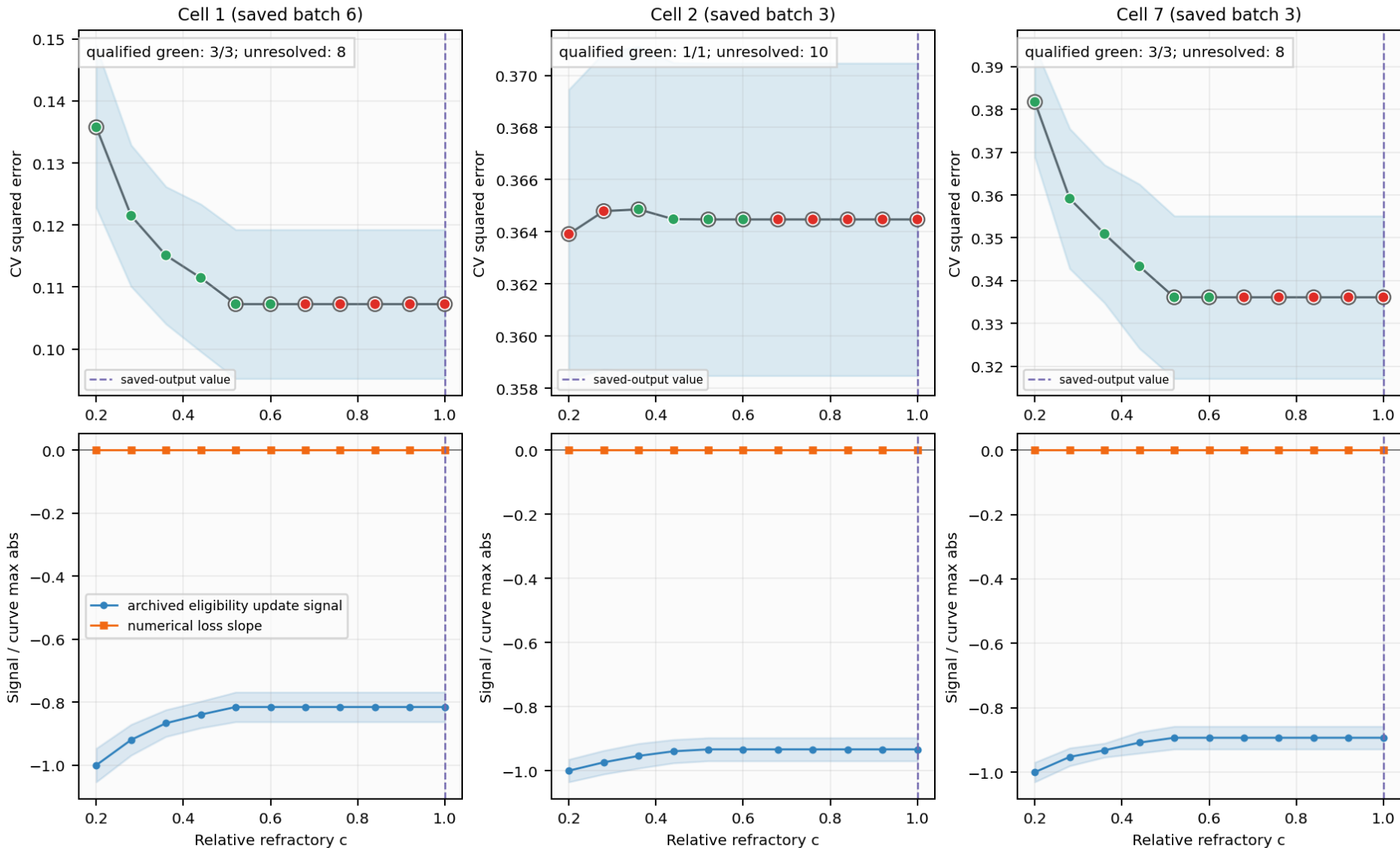
Relative refractory b: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0	0.10724	1.8590e-09	0	44.66	RED	INCONCLUSIVE	0.0%
1	0.25	0.10724	1.8590e-09	0	44.66	RED	INCONCLUSIVE	100.0%
1	0.5	0.10724	3.7180e-09	1.1565e-17	44.66	GREEN	INCONCLUSIVE	100.0%
1	1	0.10724	1.1154e-08	5.7824e-18	44.66	GREEN	INCONCLUSIVE	100.0%
1	2	0.10724	7.8079e-08	2.8912e-18	44.66	GREEN	INCONCLUSIVE	100.0%
1	4	0.10724	4.2590e-06	0	44.66	RED	INCONCLUSIVE	100.0%
1	6	0.10724	2.3081e-04	0	44.66	RED	INCONCLUSIVE	100.0%
1	8	0.10724	0.008765	0	44.66	RED	INCONCLUSIVE	100.0%
1	10	0.10724	0.004725	-2.6021e-18	44.66	RED	INCONCLUSIVE	100.0%
1	15	0.10724	0.004491	4.3368e-19	44.66	GREEN	INCONCLUSIVE	100.0%
1	25	0.10724	0.7202	0	44.66	RED	INCONCLUSIVE	0.0%
2	0	0.36448	0	0	17.61	RED	INCONCLUSIVE	0.0%
2	0.25	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	0.5	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	1	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	2	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	4	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	6	0.36448	0	0	17.61	RED	INCONCLUSIVE	100.0%
2	8	0.36448	0	-7.4015e-17	17.61	RED	INCONCLUSIVE	100.0%
2	8.555	0.36448	0	-7.8641e-17	17.61	RED	INCONCLUSIVE	100.0%
2	10	0.36448	0	2.3130e-17	17.61	RED	INCONCLUSIVE	100.0%
2	15	0.36448	0	1.1565e-18	17.61	RED	INCONCLUSIVE	100.0%
2	25	0.36448	0	0	17.61	RED	INCONCLUSIVE	0.0%
7	0	0.33612	2.4344e-09	0	15.06	RED	INCONCLUSIVE	0.0%
7	0.25	0.33612	4.0574e-09	0	15.06	RED	INCONCLUSIVE	100.0%
7	0.5	0.33612	6.4918e-09	9.2519e-18	15.06	GREEN	INCONCLUSIVE	100.0%
7	1	0.33612	1.7041e-08	0	15.06	RED	INCONCLUSIVE	100.0%
7	1.1437	0.33612	2.2721e-08	-1.8272e-16	15.06	RED	INCONCLUSIVE	100.0%
7	2	0.33612	1.2659e-07	9.2519e-18	15.06	GREEN	INCONCLUSIVE	100.0%
7	4	0.33612	6.8919e-06	0	15.06	RED	INCONCLUSIVE	100.0%
7	6	0.33612	3.6625e-04	0	15.06	RED	INCONCLUSIVE	100.0%
7	8	0.33612	0.006674	0	15.06	RED	INCONCLUSIVE	100.0%
7	10	0.33612	6.3060e-04	-3.4694e-18	15.06	RED	INCONCLUSIVE	100.0%
7	15	0.33612	0.01623	3.4694e-18	15.06	GREEN	INCONCLUSIVE	100.0%
7	25	0.33612	0.5744	0	15.06	RED	INCONCLUSIVE	0.0%

Relative refractory c

Objective: CV squared error. Qualified stable green/red: 7/0; qualified green fraction 100.0%. Inconclusive: 26; invalid CV: 0; raw green/all: 16/33.

Relative refractory c: archived local eligibility versus sampled loss slope



Relative refractory c: numerical values

Cell	Parameter value	Loss mean	Update signal	Sampled slope	FR (Hz)	Raw color	Qualified status	CV stencil valid
1	0.2	0.13576	-1.308e+04	-0.1784	43.82	GREEN	INCONCLUSIVE	0.0%
1	0.28	0.12149	-1.203e+04	-0.1291	44.15	GREEN	STABLE GREEN	100.0%
1	0.36	0.1151	-1.133e+04	-0.0626	44.47	GREEN	STABLE GREEN	100.0%
1	0.44	0.11147	-1.098e+04	-0.04915	44.58	GREEN	STABLE GREEN	100.0%
1	0.52	0.10724	-1.066e+04	-0.02645	44.66	GREEN	INCONCLUSIVE	100.0%
1	0.6	0.10724	-1.066e+04	-7.4015e-17	44.66	GREEN	INCONCLUSIVE	100.0%
1	0.68	0.10724	-1.066e+04	0	44.66	RED	INCONCLUSIVE	100.0%
1	0.76	0.10724	-1.066e+04	0	44.66	RED	INCONCLUSIVE	100.0%
1	0.84	0.10724	-1.066e+04	7.4015e-17	44.66	RED	INCONCLUSIVE	100.0%
1	0.92	0.10724	-1.066e+04	0	44.66	RED	INCONCLUSIVE	100.0%
1	1	0.10724	-1.066e+04	0	44.66	RED	INCONCLUSIVE	0.0%
2	0.2	0.36392	-1.571e+04	0.01097	17.58	RED	INCONCLUSIVE	0.0%
2	0.28	0.36479	-1.53e+04	0.005911	17.59	RED	INCONCLUSIVE	100.0%
2	0.36	0.36486	-1.498e+04	-0.001931	17.61	GREEN	INCONCLUSIVE	100.0%
2	0.44	0.36449	-1.476e+04	-0.002417	17.61	GREEN	STABLE GREEN	100.0%
2	0.52	0.36448	-1.466e+04	-6.0163e-05	17.61	GREEN	INCONCLUSIVE	100.0%
2	0.6	0.36448	-1.466e+04	-2.9606e-16	17.61	GREEN	INCONCLUSIVE	100.0%
2	0.68	0.36448	-1.466e+04	0	17.61	RED	INCONCLUSIVE	100.0%
2	0.76	0.36448	-1.466e+04	0	17.61	RED	INCONCLUSIVE	100.0%
2	0.84	0.36448	-1.466e+04	1.4803e-16	17.61	RED	INCONCLUSIVE	100.0%
2	0.92	0.36448	-1.466e+04	1.4803e-16	17.61	RED	INCONCLUSIVE	100.0%
2	1	0.36448	-1.466e+04	0	17.61	RED	INCONCLUSIVE	0.0%
7	0.2	0.38172	-7700	-0.2821	14.54	GREEN	INCONCLUSIVE	0.0%
7	0.28	0.35915	-7336	-0.1926	14.85	GREEN	STABLE GREEN	100.0%
7	0.36	0.35091	-7177	-0.09882	14.92	GREEN	STABLE GREEN	100.0%
7	0.44	0.34334	-6994	-0.09243	15.02	GREEN	STABLE GREEN	100.0%
7	0.52	0.33612	-6878	-0.04515	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.6	0.33612	-6878	-2.2204e-16	15.06	GREEN	INCONCLUSIVE	100.0%
7	0.68	0.33612	-6878	0	15.06	RED	INCONCLUSIVE	100.0%
7	0.76	0.33612	-6878	0	15.06	RED	INCONCLUSIVE	100.0%
7	0.84	0.33612	-6878	1.4803e-16	15.06	RED	INCONCLUSIVE	100.0%
7	0.92	0.33612	-6878	2.2204e-16	15.06	RED	INCONCLUSIVE	100.0%
7	1	0.33612	-6878	0	15.06	RED	INCONCLUSIVE	0.0%

Limitations and what this test can establish

Sampled hard-spike loss

The forward model contains threshold and stochastic spike events, so the loss is a staircase-like sampled function rather than a globally smooth curve. Common random numbers reduce Monte Carlo noise but do not make the hard-event map differentiable.

Consistency check, not derivative validation

A stable-green point means the archived update signal has the same direction as the coarse sampled curve under the strict descriptive rule used here. It does not establish an exact derivative or magnitude, nor guarantee that a finite Adam step will lower loss after momentum, normalization, clipping, or parameter interactions.

One-dimensional slices

Each curve holds the other parameters fixed. It can expose a wrong local direction, flat regions, and boundaries, but it cannot detect compensating parameter combinations or establish joint identifiability of PV/E interaction parameters.

Only three cells and 3 seeds

The test intentionally uses cells 1, 2, and 7 and 3 independent stochastic seeds to keep a full 12-parameter analysis tractable. Broader biological claims would require more cells, checkpoint basins, seeds, and held-out stimuli.

Historical loss split

The refractory update signals are attached to CV squared error in this archived code, whereas all other update signals are attached to PSTH SSE. Comparing a refractory signal to PSTH loss would answer the wrong question for this implementation.

Refractory signal is heuristic

The archived refractory update is proportional to $2(\text{CV}_{\text{sim}} - \text{CV}_{\text{data}})$ times a refractory eligibility accumulator. It does not explicitly carry the derivative of CV through ISIs and event times. Therefore even a stable-green CV point is evidence of directional consistency for this heuristic, not validation of the exact CV derivative.

Archived bin convention preserved

At a PSTH boundary the archived code updates eligibility at the closing timestep but counts spikes from the preceding interval, excluding that closing timestep. This report preserves that implementation exactly instead of silently fixing it.

Sparse CV convention

When too few simulated interspike intervals exist, the archived refractory objective contributes zero loss and zero update signal. This report marks an interior point invalid when its center or either neighboring CV is undefined and excludes it from the qualified percentage.

Coarse adjacent-point slopes

The broad grids are intended to reveal loss-landscape shape. Their adjacent-point slopes are coarse secant diagnostics, especially for logarithmic STRF and conductance grids; they are not converged local finite differences. Endpoints and turning/flat secants are therefore excluded from the qualified result.

Constrained boundaries

Several selected solutions are at allowed boundaries, including some conductances, absolute refractory periods, relative refractory b values, and relative refractory c values. Boundary points are endpoints of these slices and are excluded from the two-sided qualified comparison.

In-sample batch selection

Batches were selected using PSTH SSE on the same recording analyzed here. Neither the selected basin nor the directional results are a held-out generalization test.

Files accompanying this PDF: raw_results.csv contains every seed/run/cell observation; per_seed_slopes.csv contains adjacent-point slopes and qualification fields before seed averaging; aggregate_results.csv contains plotted values, raw colors, and qualified statuses; alignment_summary.csv contains the report summary; metadata.json contains provenance, grids, baseline values, and classification rules.